Week 4 – Forward Kinematics

ELEC0129 Introduction to Robotics

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Schedule

Legend:
A: MPEB 6.02 Teaching Lab
B: Roberts 7.04 PC Lab

Week	Recorded, uploaded by Friday of previous week	Workshop, Mondays 11am-1pm and/or Wednesdays 9am-11am	Notes
1	(Scenario Week)		
2	Lec: Intro; Spatial description	A: Workshop: Offline programming	Half class Mon, half class Wed
3	Lec & Tut: Spatial description	A: Workshop: Build robot	
4	Lec & Tut: Forward kinematics	A: Workshop: Forward kinematics	
5	Lec & Tut: Inverse kinematics	B: Workshop: Offline programming	Half class Mon, half class Wed
RW	(Reading Week)		
6	(Scenario Week)		
7	Lec & Tut: Jacobians	A: Workshop: Inverse kinematics	
8	Lec: Trajectory Planning	A: Workshop: Trajectory planning	
9	Lec & Tut: Dynamics	A: Workshop: Trajectory planning	
10	Lec & Tut: Control	A: Workshop: Pick-and-place demo	

Content

- Introduction to Forward Kinematics
- Denavit-Hartenberg Parameters
- One Complete Example
- More Examples
- Tutorial Questions

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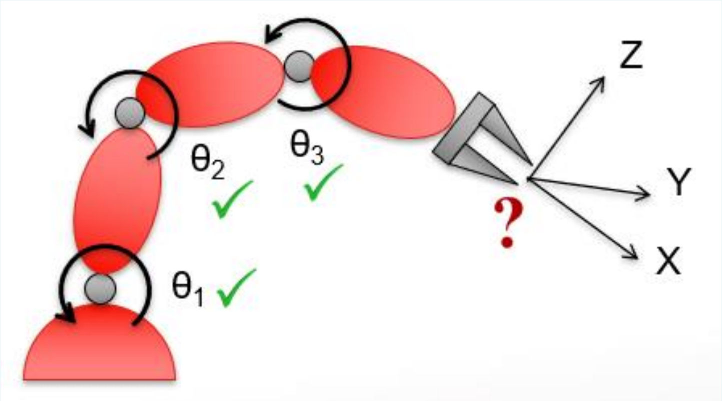
Introduction (1)

- Kinematics is the study of motion without regard to forces which causes it.
- Of interests are position, velocity, acceleration and higher order derivatives.

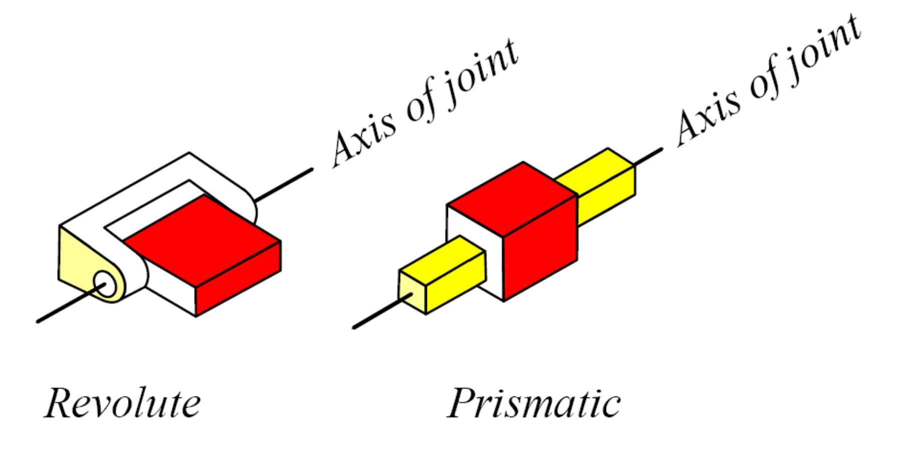
Introduction (2)

- In this lecture, position and orientation in static situations will be discussed:
  - Given the joint space parameters (angles for revolute joints, or offsets for prismatic joints); and
  - Given the dimensions of the links;
  - What is the position and orientation of the end-effector in Cartesian space?



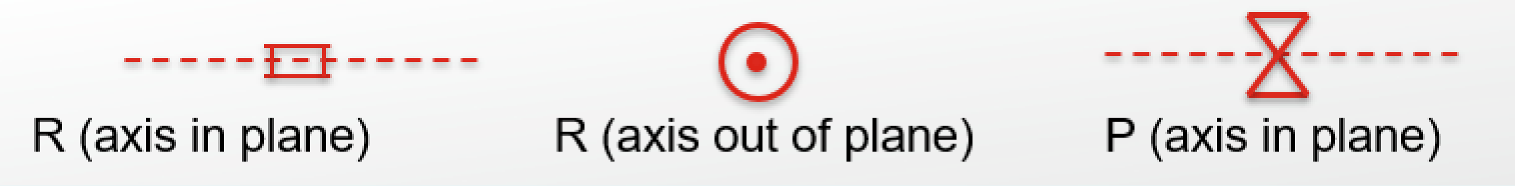
Recall: Robot Joints (1)

- Industrial robots are mostly made by links connected by joints.
  - The joints could be revolute (R) or prismatic (P).



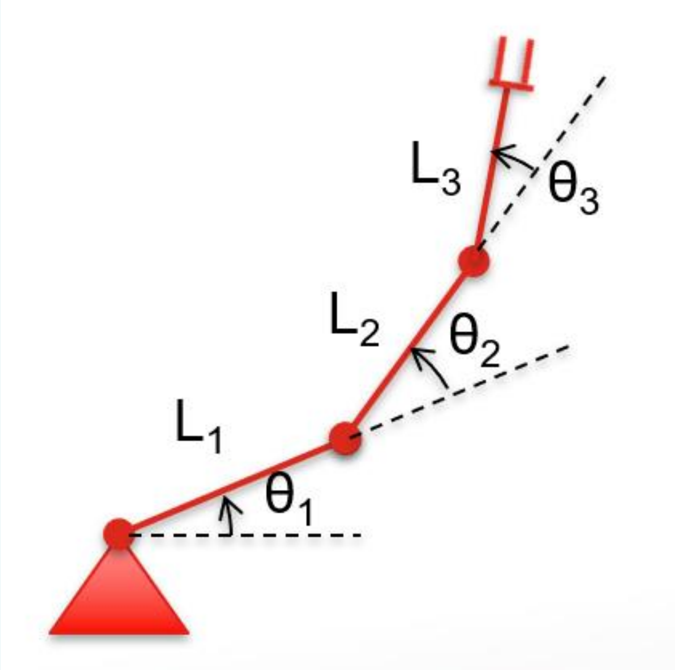
Recall: Robot Joints (2)

- The 2D schematics of the joints are as follows:



A Simple Example (1)

- Consider a 3-link planar robot as shown on the right.
  - The link lengths and the joint angles are known.
  - What are the position and orientation of the end-effector?

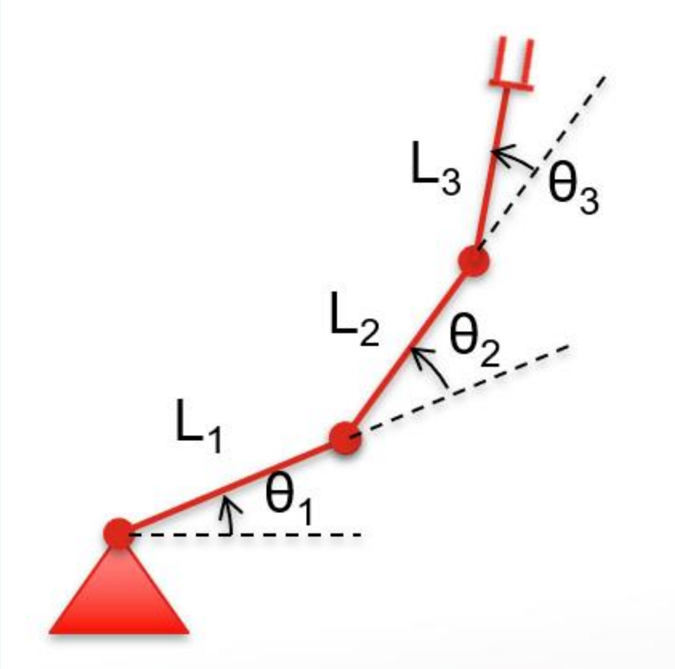


A Simple Example (2)

- In this simple example, the position can be calculated based on trigonometry:

$$
\begin{aligned}
x = {} & L_1 \cos\theta_1 \\
& + L_2 \cos(\theta_1 + \theta_2) \\
& + L_3 \cos(\theta_1 + \theta_2 + \theta_3)
\end{aligned}
$$

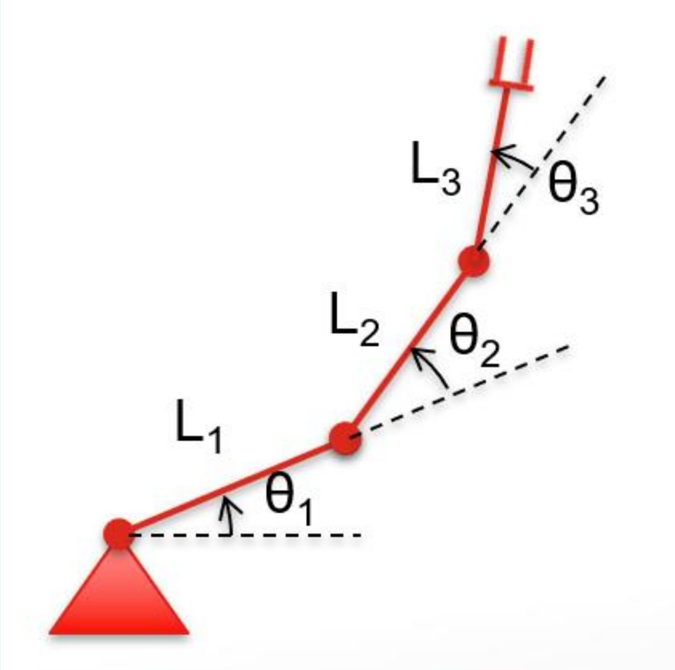
$$
\begin{aligned}
y = {} & L_1 \sin\theta_1 \\
& + L_2 \sin(\theta_1 + \theta_2) \\
& + L_3 \sin(\theta_1 + \theta_2 + \theta_3)
\end{aligned}
$$



A Simple Example (3)

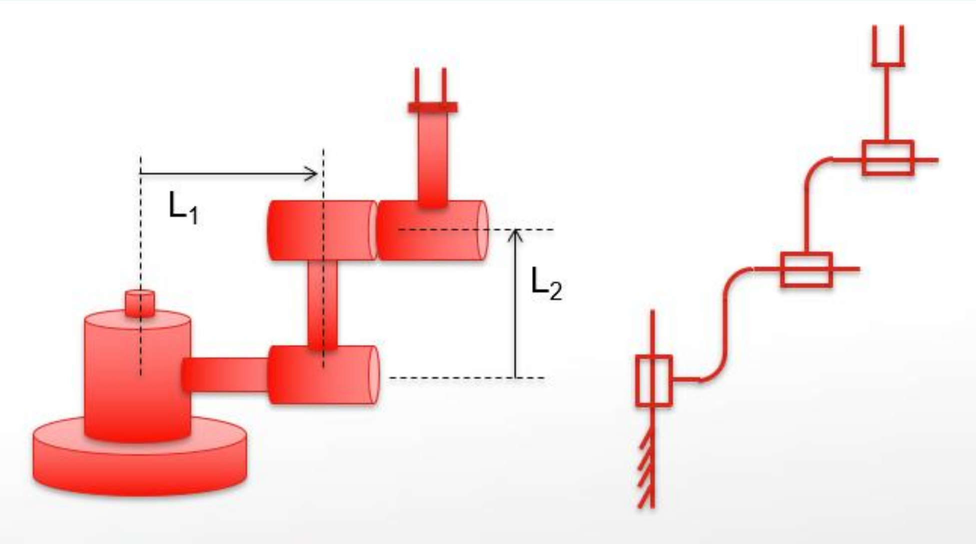
- The orientation is also straightforward:

$$\boxed{\theta_{\text{total}} = \theta_1 + \theta_2 + \theta_3}$$



For More General Cases (1)

- For more general cases, especially if the robot is non-planar, it will not be so easy to visualize the answer.



For More General Cases (2)

- We need a more systematic way to solve the problem:
  - Attach frames to various parts of mechanism.
  - Find the relationships between neighbouring frames:

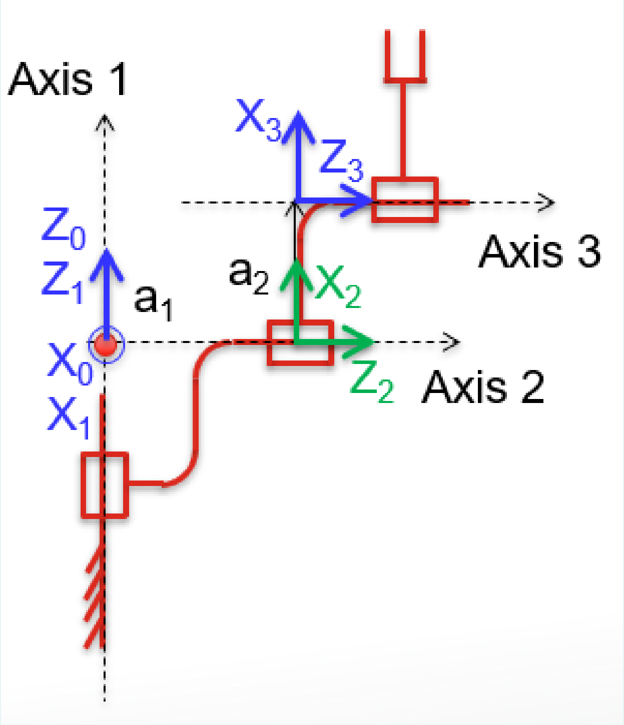
$$\boxed{{}^{0}_{1}T, {}^{1}_{2}T, \cdots, {}^{n-1}_{n}T}$$

  - Use compound transform to calculate the position of the last frame.

$$\boxed{{}^{0}_{n}T = {}^{0}_{1}T \cdot {}^{1}_{2}T \cdot \cdots \cdot {}^{n-1}_{n}T}$$

  - With ${}^{n}P$ known, calculate:

$$\boxed{{}^{0}P = {}^{0}_{n}T \cdot {}^{n}P}$$



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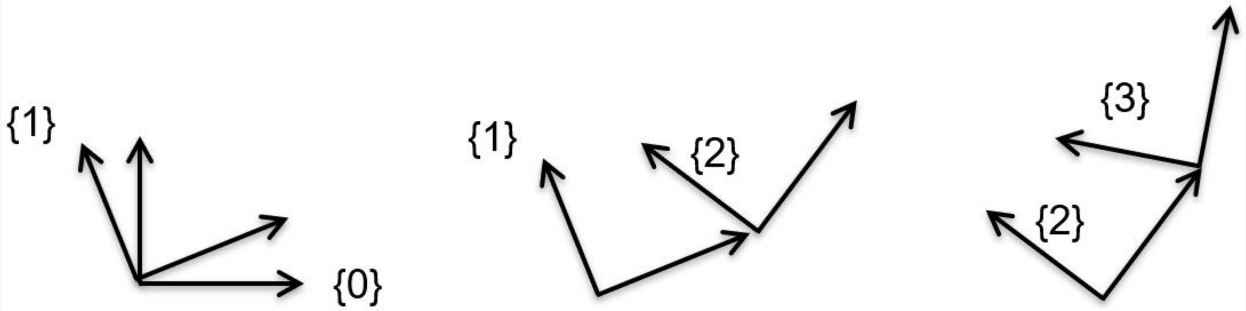
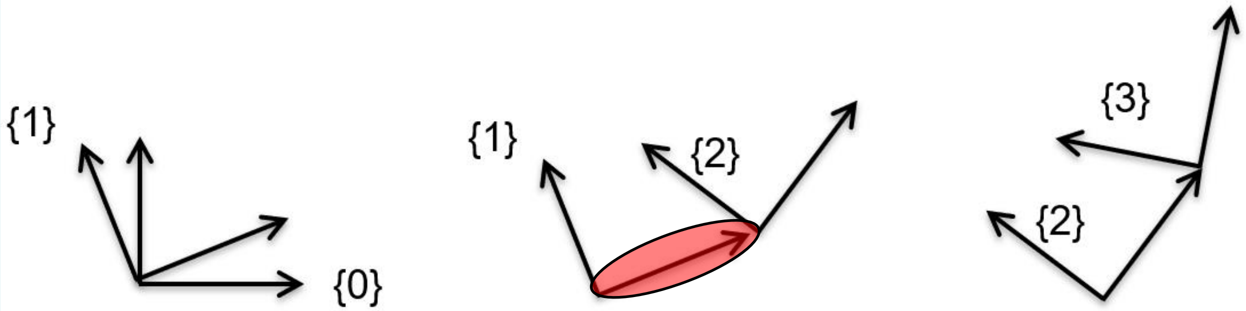
Denavit-Hartenberg Parameters (1)

- We mentioned we will need transformation between neighbouring frames: ${}^{0}_{1}T, {}^{1}_{2}T, \cdots, {}^{n-1}_{n}T$
- Question: How many parameters are needed to describe each transformation?
  - In general transformations, six: three for position and three for orientation.



Denavit-Hartenberg Parameters (2)

- However, in robots, certain values are fixed.
  - E.g. the distance between origins of {1} and {2} if joint 1 is revolute – constrained by the link length.
  - Also, each joint is constrained to provide only 1 DOF.



Denavit-Hartenberg Parameters (3)

- Therefore, for robotic manipulators, we only need 4 parameters to describe the transformation between each links. They are:
  - Link lengths a_{i-1}.
  - Link twists α_{i-1}.
  - Joint angles θ_i.
  - Link offsets d_i.
- The above four parameters are called Denavit-Hartenberg Parameters, or in short "DH Parameters".

Denavit-Hartenberg Parameters (4)

- In the next sections, we will learn through examples:
  - The definition of DH Parameters;
  - How to attach the frames;
  - How to get the Cartesian description of the last link;
  - How to get the Cartesian description of end-effector.

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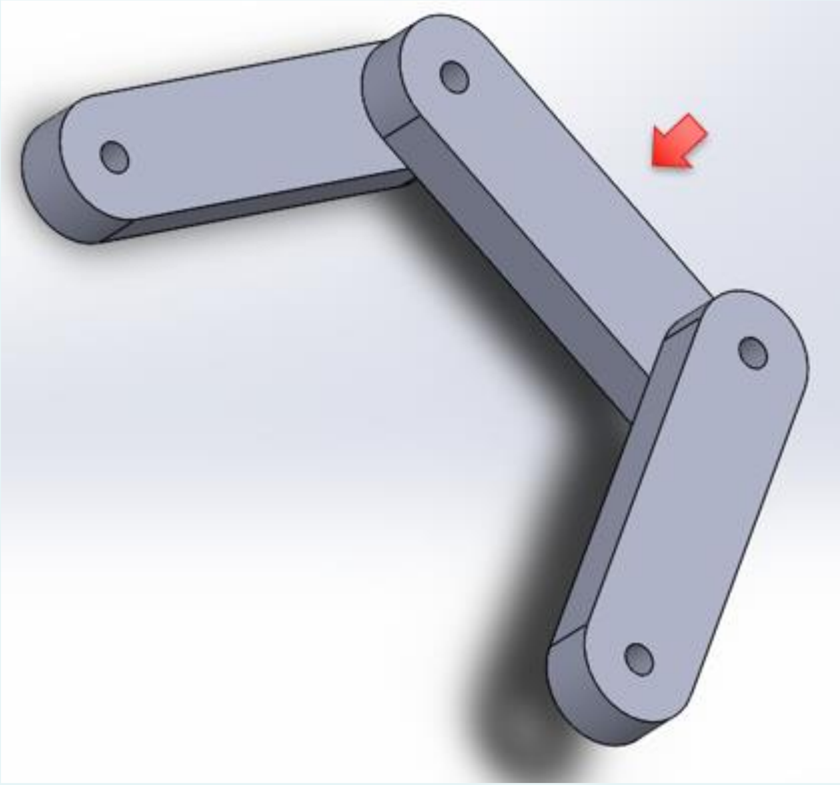
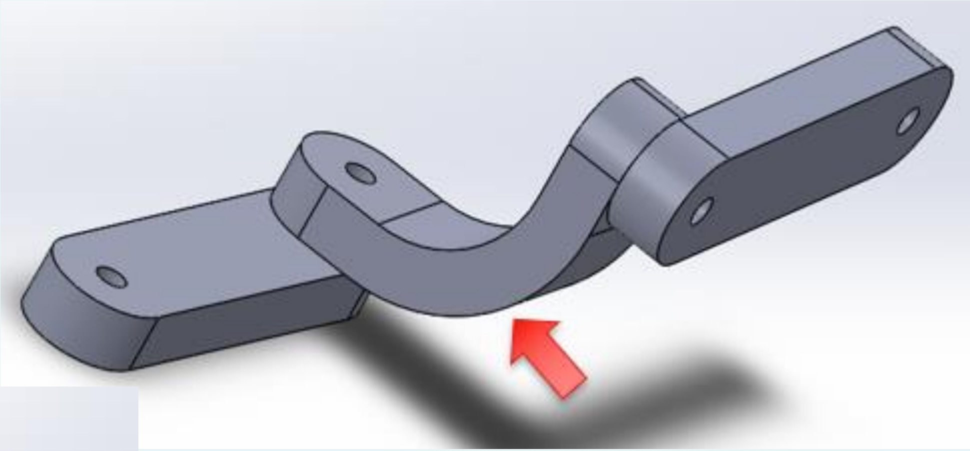
Link Length (1)

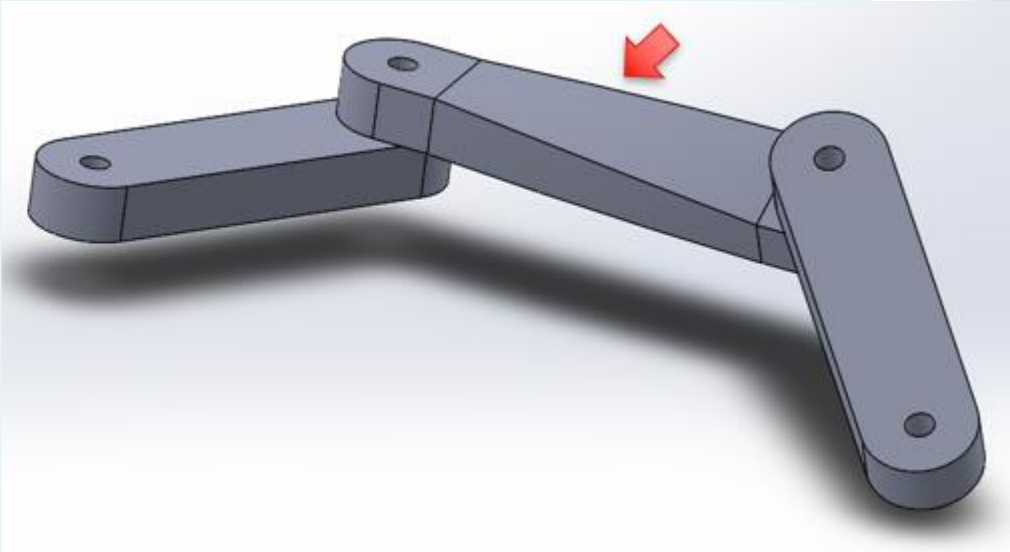
- A link is basically just a rigid body between two joint axes.
- It turns out that a link can be fully described by its length and its twist.
- What is the length of the link shown in the picture?



Link Length (2)

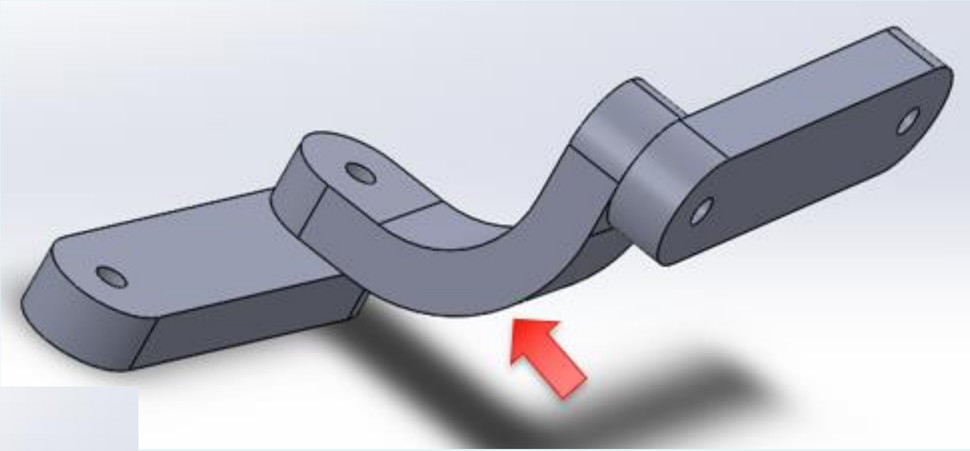
- What about these?

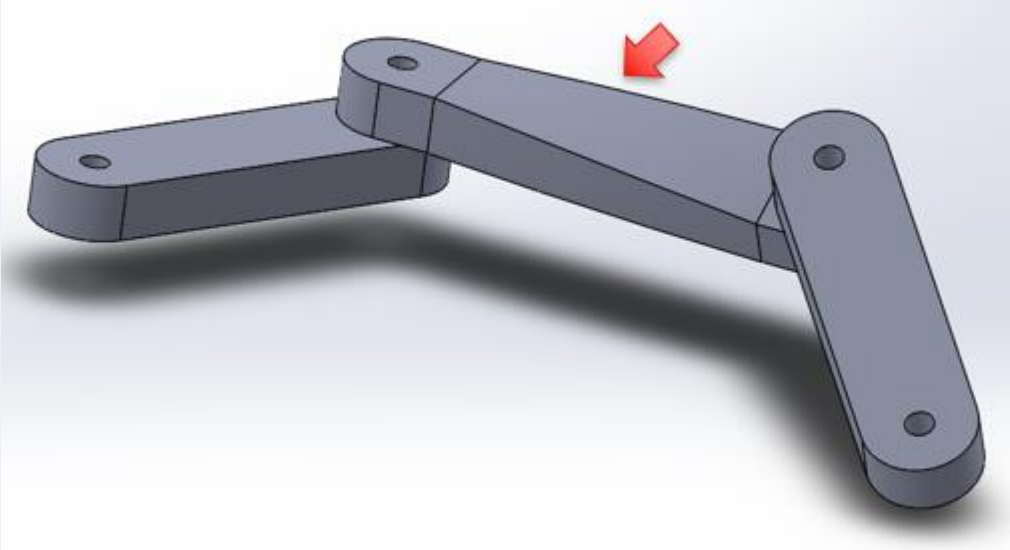




Link Twist

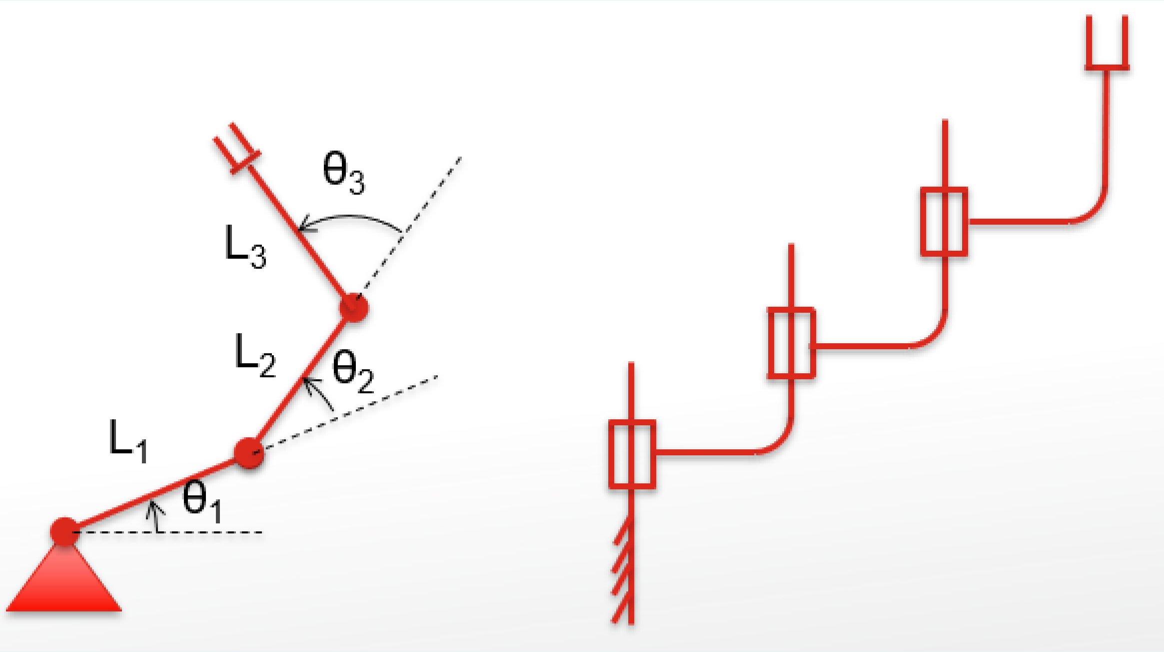
- Also, what are twists?





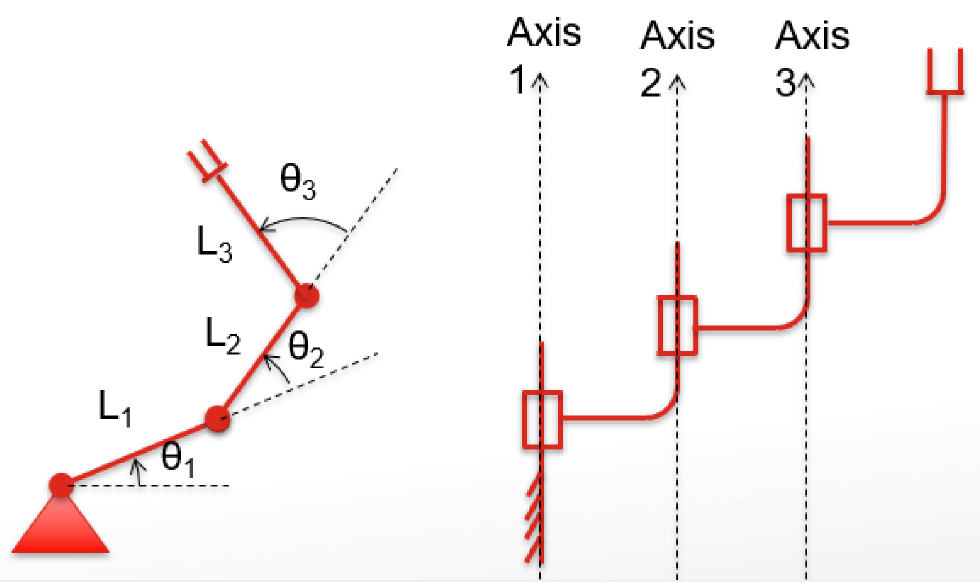
Example 1

- 3 link RRR planar robot:



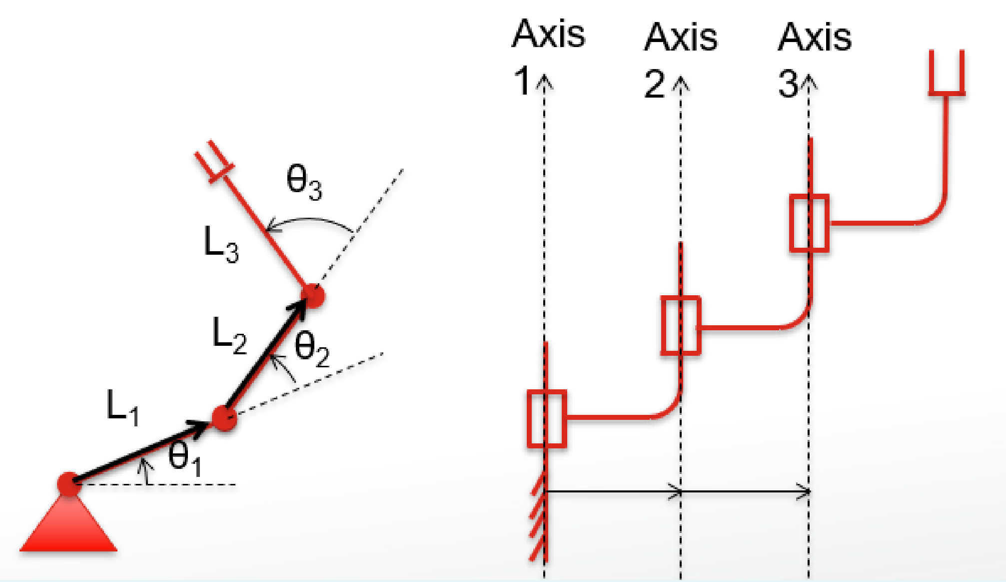
Example 1 – Step 1

- Draw axes:
  - For revolute joints: About the rotation.
  - For prismatic joints: Along the translation.
  - The direction of arrow is arbitrary, but once fixed must be followed through til the end.



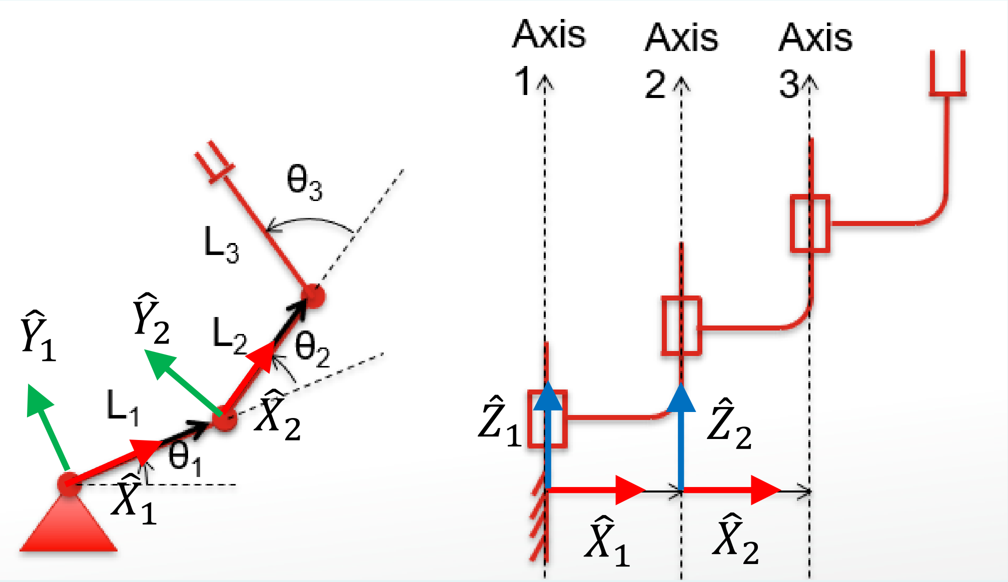
Example 1 – Step 2

- Draw the mutually perpendicular lines between axes.
  - Draw from lower to higher axis, e.g. 1 to 2, 2 to 3…
  - In this example, since the axes are parallel, the mutual perpendicular can be placed in any arbitrary position.
  - We will put them in the same plane.



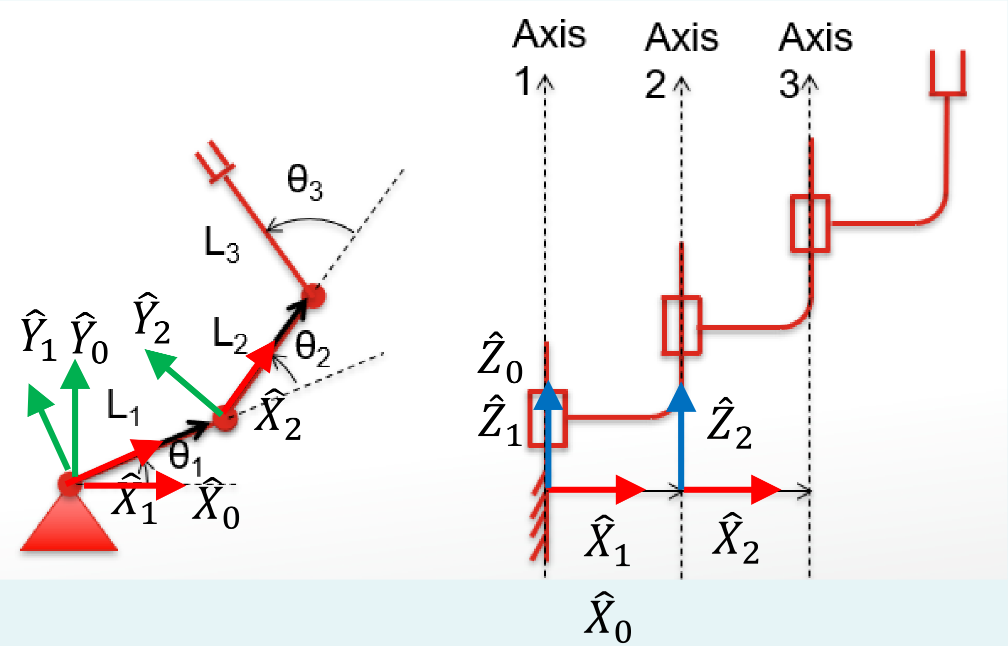
Example 1 – Step 3

- Attach frames $\{1\}$ to $\{n-1\}$ as follows:
  - $\hat{Z}_i$ pointing along the i-th joint axis.
  - $\hat{X}_i$ pointing along mutual perpendicular.
  - Assign $\hat{Y}_i$ to complete the right-hand coordinate system.



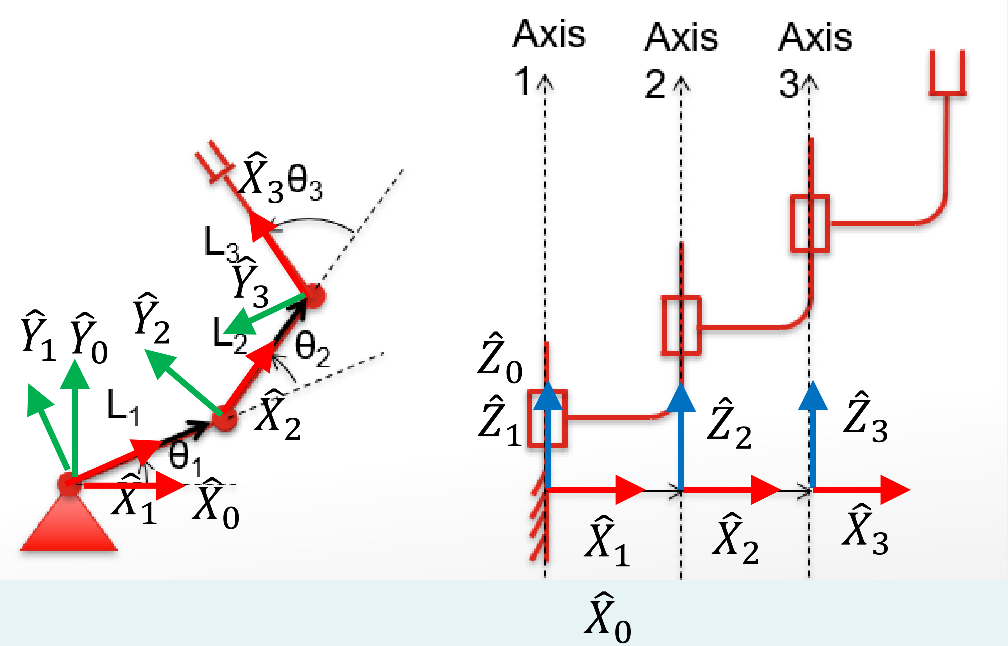
Example 1 – Step 4

- Attach frame {0}:
  - Assign {0} to match {1}
    WHEN the first joint variable
    (in this case θ_1) is zero.



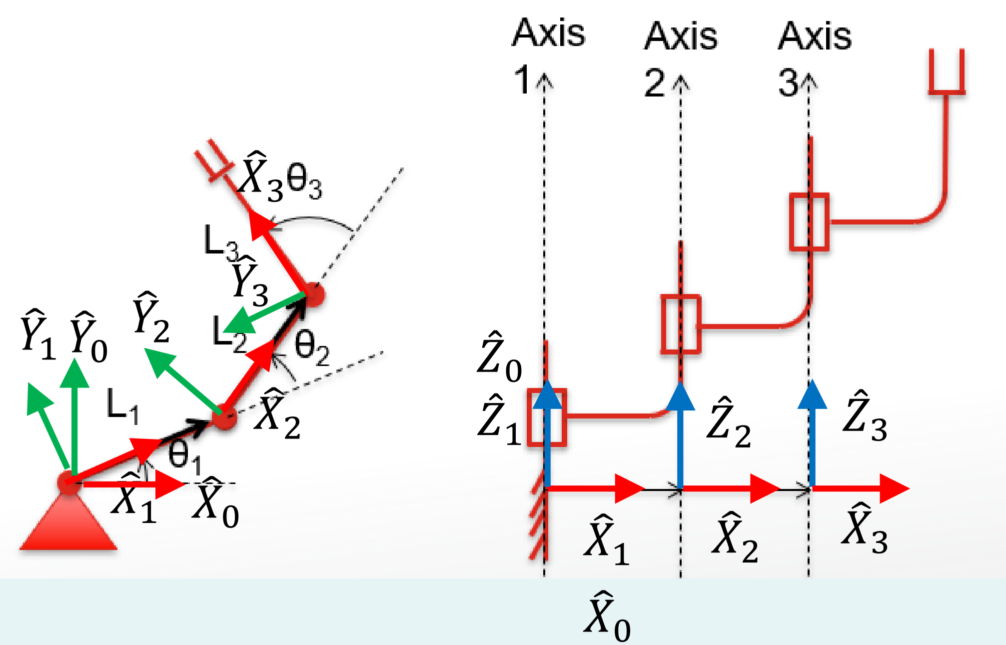
Example 1 – Step 5

- Attach frame $\{n\}$:
  - $\hat{Z}_n$ pointing along the n-th joint axis.
  - $\hat{X}_n$ pointing in the direction such that if the n-th joint parameter (in this example θ_3) is zero, then $\hat{X}_n$ is aligned with $\hat{X}_{n-1}$.



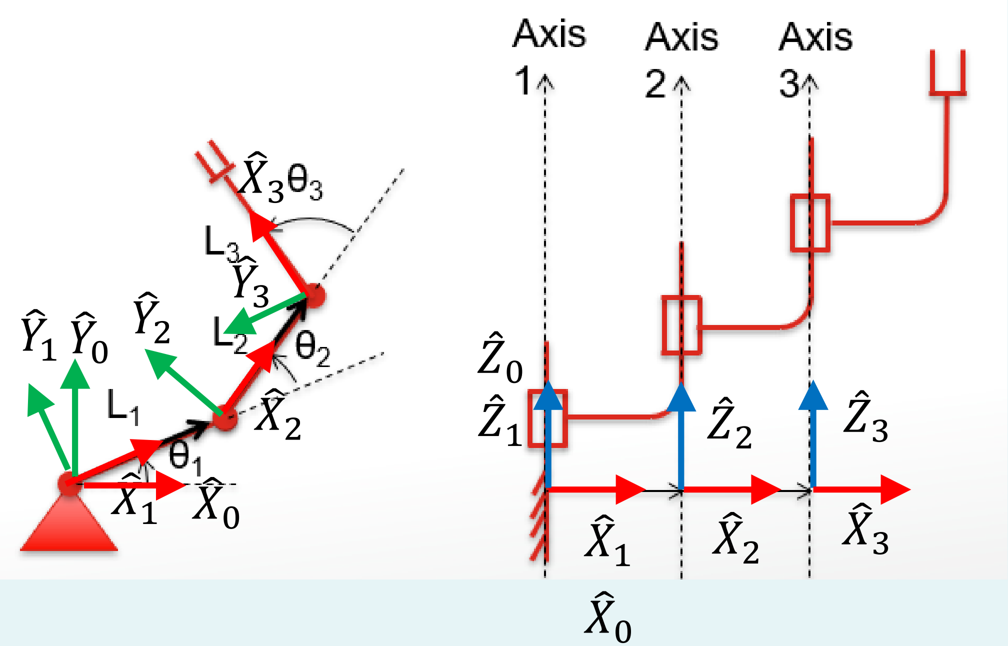
Example 1 – Step 6

- Put in the link lengths a_0 to a_{n-1}.
- Definition: a_{i-1} = distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ along $\hat{X}_{i-1}$.
  - a_0 = distance from $\hat{Z}_0$ to $\hat{Z}_1$ along $\hat{X}_0$ = 0.
  - a_1 = distance from $\hat{Z}_1$ to $\hat{Z}_2$ along $\hat{X}_1 = L_1$.
  - a_2 = distance from $\hat{Z}_2$ to $\hat{Z}_3$ along $\hat{X}_2 = L_2$.



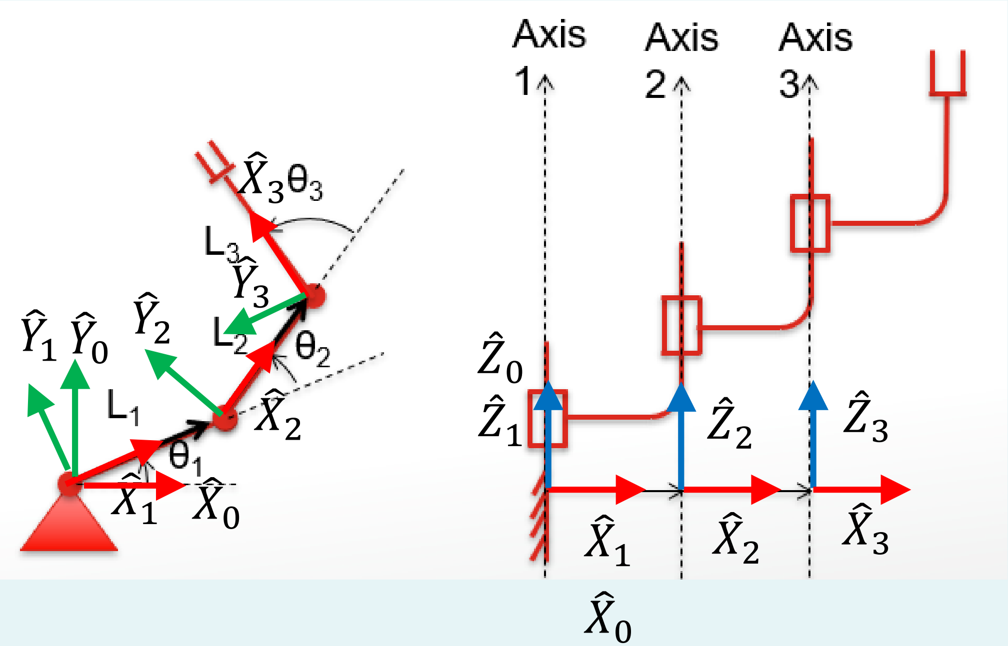
Example 1 – Step 7

- Put in the link twists α_0 to α_{n-1}.
- Definition: α_{i-1} = angle between $\hat{Z}_{i-1}$ to $\hat{Z}_i$ about $\hat{X}_{i-1}$.
  - α_0 = angle between $\hat{Z}_0$ to $\hat{Z}_1$ about $\hat{X}_0$ = 0.
  - α_1 = angle between $\hat{Z}_1$ to $\hat{Z}_2$ about $\hat{X}_1$ = 0.
  - α_2 = angle between $\hat{Z}_2$ to $\hat{Z}_3$ about $\hat{X}_2$ = 0.



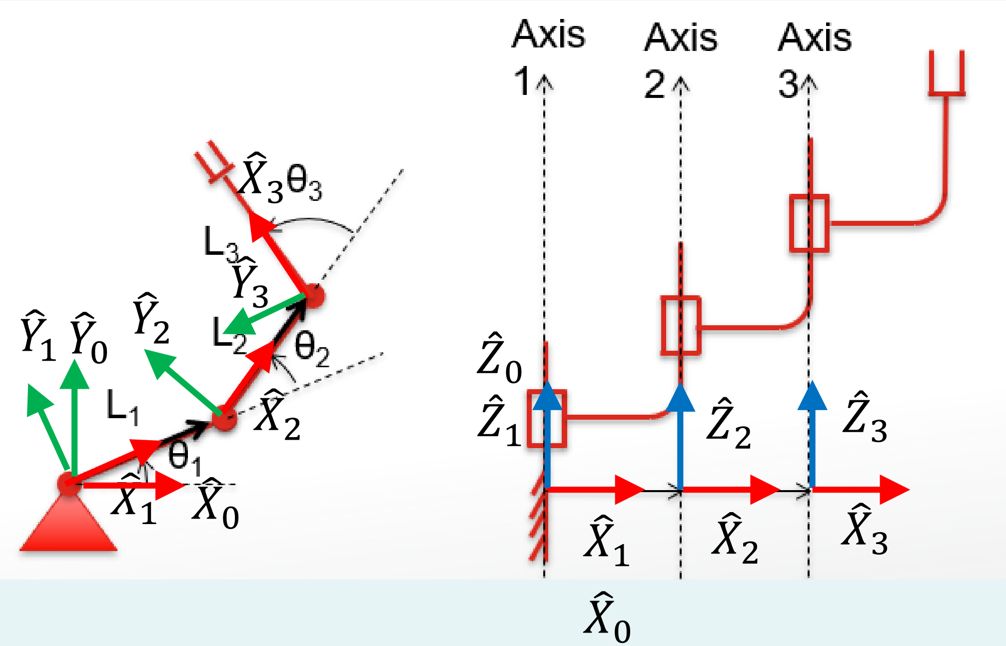
Example 1 – Step 8

- Put in the link offsets d_1 to d_n.
- Definition: d_i = distance from $\hat{X}_{i-1}$ to $\hat{X}_i$ along $\hat{Z}_i$.
  - d_1 = distance from $\hat{X}_0$ to $\hat{X}_1$ along $\hat{Z}_1$ = 0.
  - d_2 = distance from $\hat{X}_1$ to $\hat{X}_2$ along $\hat{Z}_2$ = 0.
  - d_3 = distance from $\hat{X}_2$ to $\hat{X}_3$ along $\hat{Z}_3$ = 0.



Example 1 – Step 9

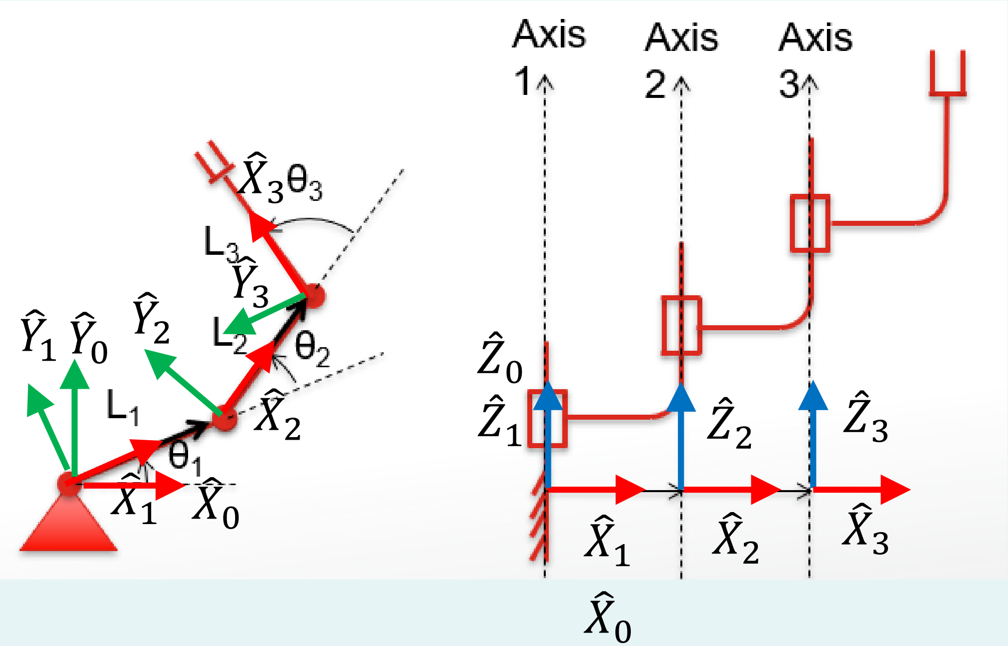
- Put in the joint angles θ_1 to θ_n.
- Definition: θ_i = angle btw. $\hat{X}_{i-1}$ to $\hat{X}_i$ about $\hat{Z}_i$.
  - θ_1 = angle btw. $\hat{X}_0$ to $\hat{X}_1$ about $\hat{Z}_1$ = variable.
  - θ_2 = angle btw. $\hat{X}_1$ to $\hat{X}_2$ about $\hat{Z}_2$ = variable.
  - θ_3 = angle btw. $\hat{X}_2$ to $\hat{X}_3$ about $\hat{Z}_3$ = variable.



Example 1 – Step 10

- Transfer the results into a DH table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



- Note: Our kinematic analysis always ends at last joint axis. Thus L_3 does not appear in the table. It is a final offset which can be added later.

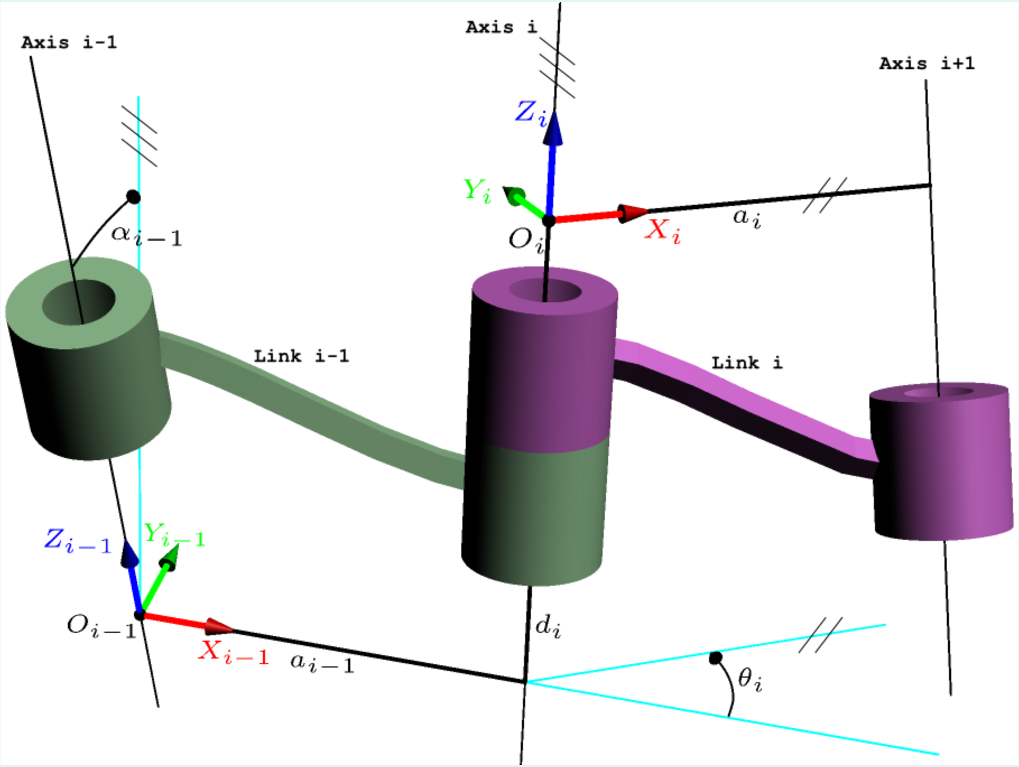
Example 1 – Halfway Summary

- Any robot can be described kinematically using the four DH parameters we have just learnt:
  - Link lengths $a_{i-1.}$
  - Link twists $\alpha_{i-1.}$
  - Joint angles $\theta_{i.}$
  - Link offsets d_i.
- a_{i-1}, α_{i-1} are always constants.
- d_i variable for prismatic joint, constant (not necessarily zero!) for revolute joint.
- θ_i variable for revolute joint, constant (not necessarily zero!) for prismatic joint.

Example 1 – Link Transformation

- Each of the transformation can be simplified to 4 sub-problems:

$$
{}^{i-1}_{i}T = R_x(\alpha_{i-1}) \cdot D_x(a_{i-1}) \cdot R_z(\theta_i) \cdot D_z(d_i)
$$



Example 1 – Link Transformation Cont'd

$$
{}^{i-1}_{\ \ i}T = R_x(\alpha_{i-1}) \cdot D_x(a_{i-1}) \cdot R_z(\theta_i) \cdot D_z(d_i)
$$

$$
= \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & c\alpha_{i-1} & -s\alpha_{i-1} & 0 \\ 0 & s\alpha_{i-1} & c\alpha_{i-1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
\boxed{{}^{i-1}_{\ \ i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}}
$$

Example 1 – Step 11

- Calculate the transformations.

$$ {}^{i-1}_{i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3

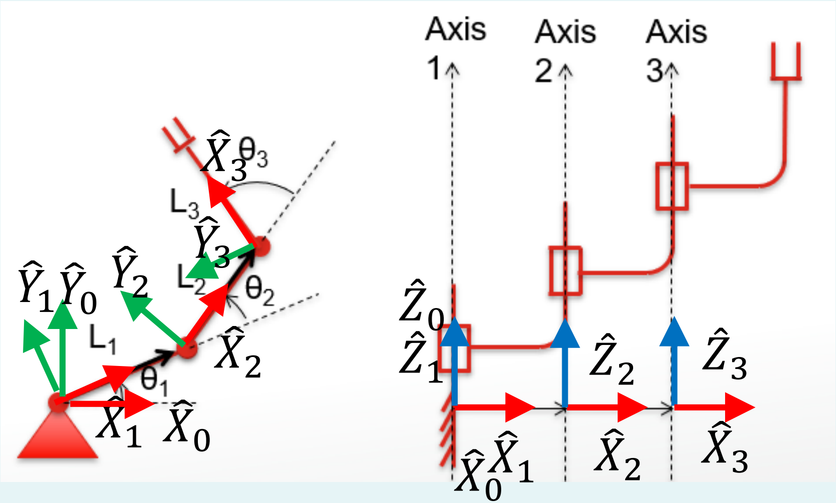
$$ {}^{0}_{1}T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{1}_{2}T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{2}_{3}T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. $$

$$ {}^{0}_{3}T = {}^{0}_{1}T \cdot {}^{1}_{2}T \cdot {}^{2}_{3}T = \begin{bmatrix} c\theta_{123} & -s\theta_{123} & 0 & L_2c\theta_{12} + L_1c\theta_1 \\ s\theta_{123} & c\theta_{123} & 0 & L_2s\theta_{12} + L_1s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

where
$$ \theta_{12} = \theta_1 + \theta_2 $$
$$ \theta_{123} = \theta_1 + \theta_2 + \theta_3 $$

Example 1 – Step 12

- By now, we have already obtained the position and orientation of frame {3}. What about the end-effector?
- According to the figure, it is at ${}^{3}P = [L_3 \quad 0 \quad 0]^T$. This is a constant! The end-effector doesn't move wrt. frame {3}!



Example 1 – Step 13

- Calculate the position and orientation of end-effector wrt. frame {0}.

$$
{}^{0}P = {}^{0}_{3}T \cdot {}^{3}P = \begin{bmatrix} c\theta_{123} & -s\theta_{123} & 0 & L_2 c\theta_{12} + L_1 c\theta_1 \\ s\theta_{123} & c\theta_{123} & 0 & L_2 s\theta_{12} + L_1 s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} L_3 \\ 0 \\ 0 \\ 1 \end{bmatrix}
= \begin{bmatrix} L_3 c\theta_{123} + L_2 c\theta_{12} + L_1 c\theta_1 \\ L_3 s\theta_{123} + L_2 s\theta_{12} + L_1 s\theta_1 \\ 0 \\ 1 \end{bmatrix}
$$

- Compare this with the results obtained through trigonometry. They are the same, but this systematic approach is useful for more general cases.

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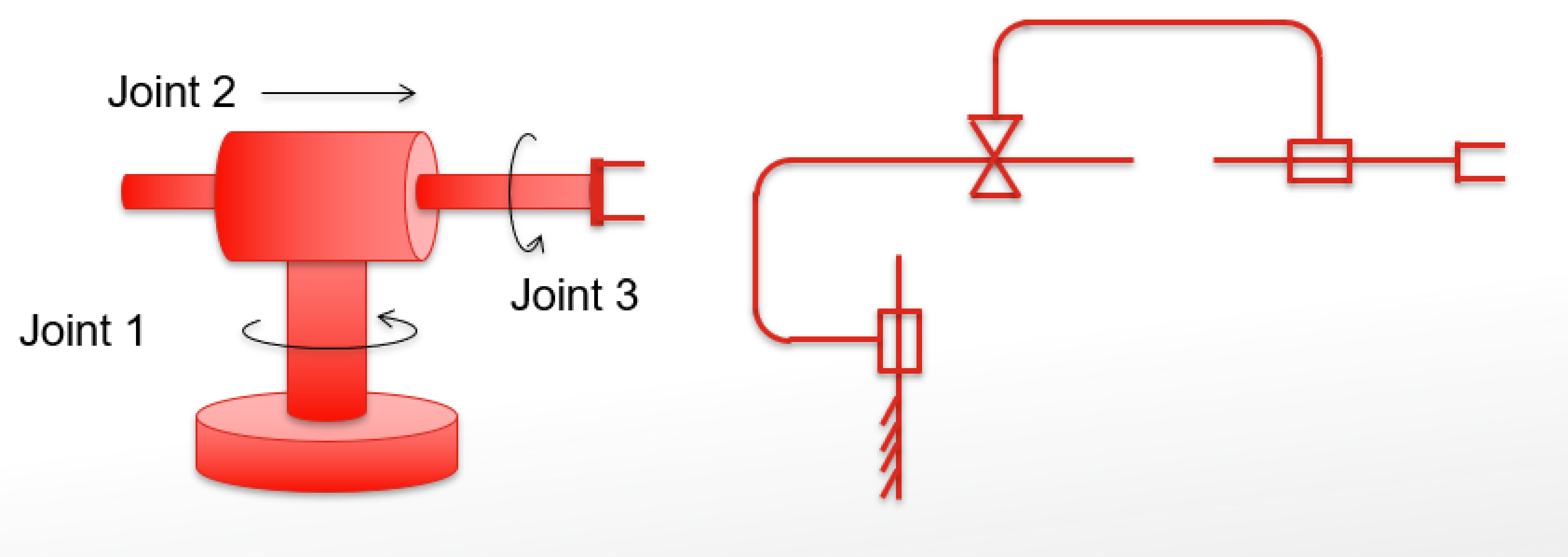
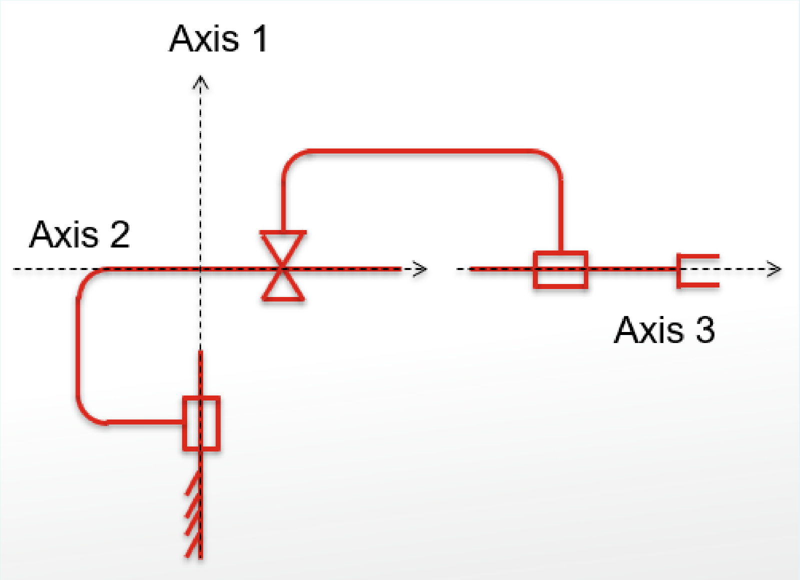
Example 2

- 3 link RPR robot:



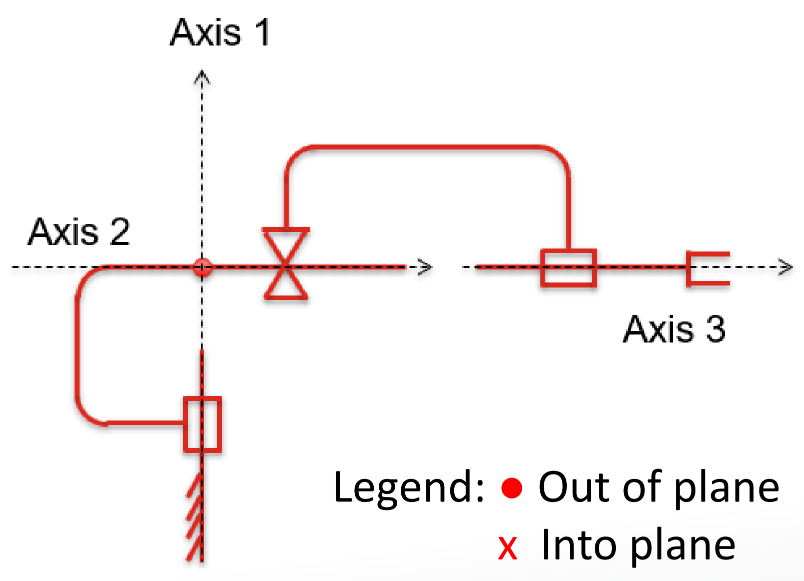
Example 2 – Step 1

- Draw axes:
  - For revolute joints: About the rotation.
  - For prismatic joints: Along the translation.
  - The direction of arrow is arbitrary, but once fixed must be followed through til the end.



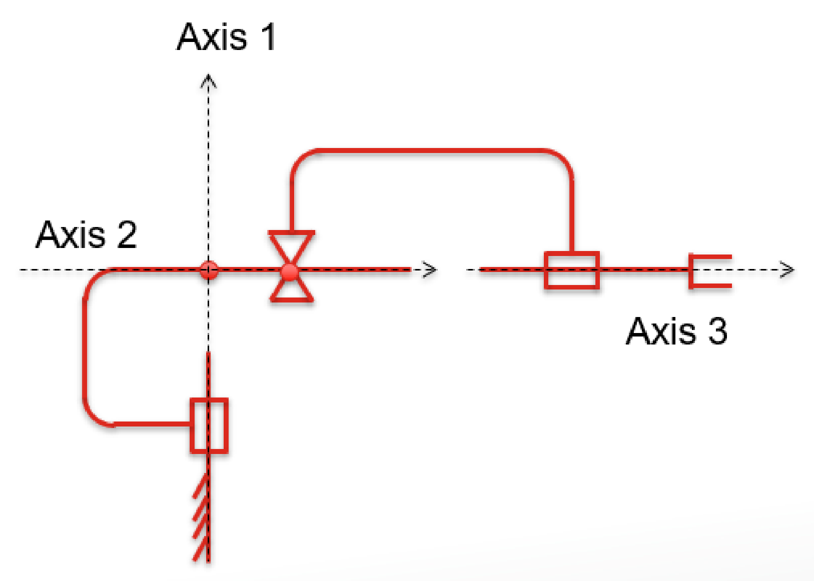
Example 2 – Step 2

- Draw the mutual perpen-dicular lines between axes.
  - Draw from lower to higher axis, e.g. 1 to 2, 2 to 3…
  - In this example, Axis 1 and Axis 2 intersect.
    - Line is perpendicular to the plane made by axes 1 and 2.
    - Because the axes intersect, the sense of direction is arbitrary (either into or out of the plane). Here I have chosen out of the plane.



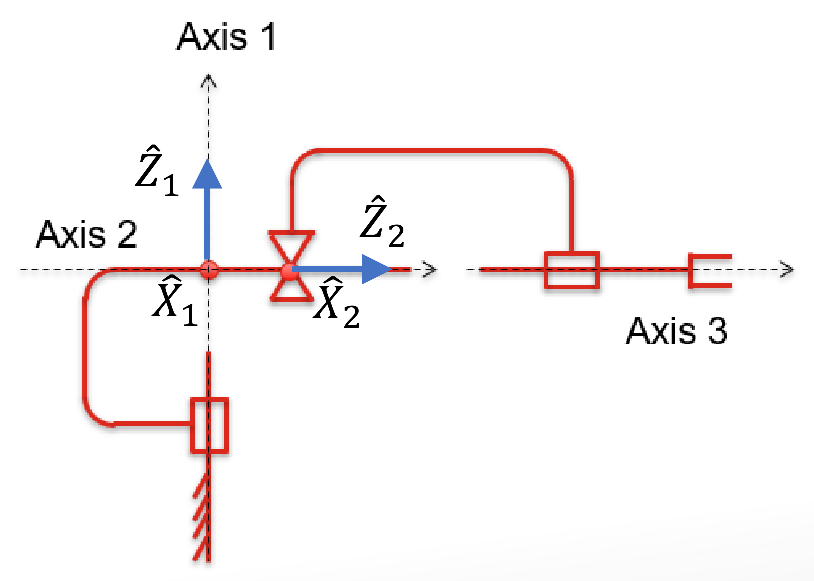
Example 2 – Step 2 Continued

- Also, Axis 2 and Axis 3 intersect (and are the same).
  - Line is arbitrary. Put it at joint 2, also out of the plane.



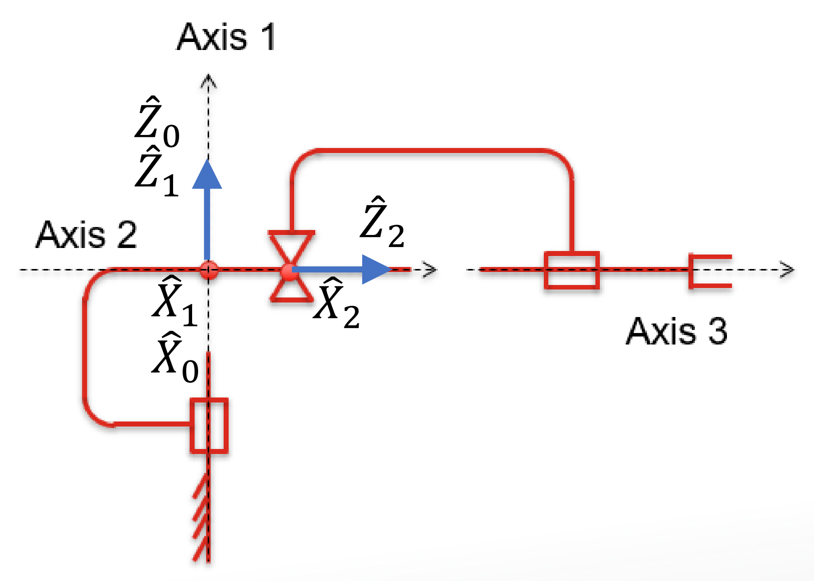
Example 2 – Step 3

- Attach frames $\{1\}$ to $\{n-1\}$ as follows:
  - $\hat{Z}_i$ pointing along the i-th joint axis.
  - $\hat{X}_i$ pointing along mutual perpendicular.
  - (Assign $\hat{Y}_i$ to complete the right-hand coordinate system.)



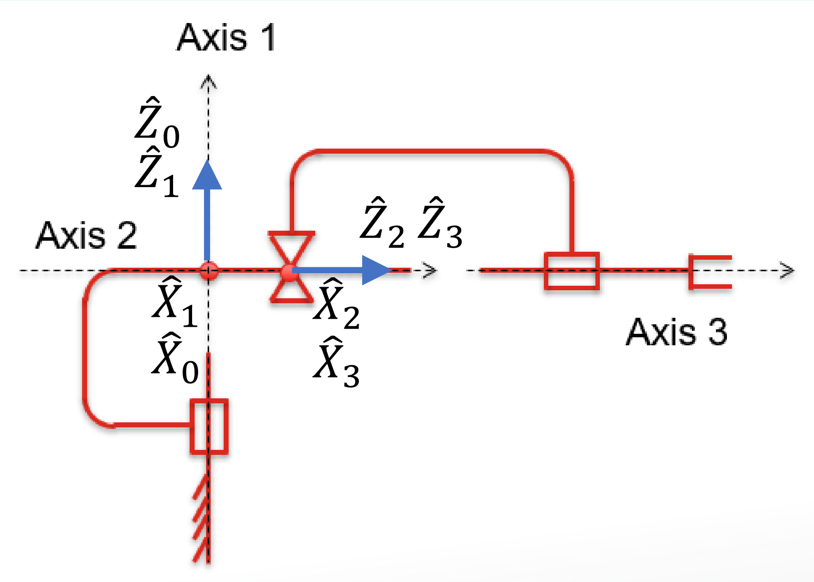
Example 2 – Step 4

- Attach frame {0}:
  - Assign {0} to match {1} WHEN the first joint variable (in this case θ_1) is zero.



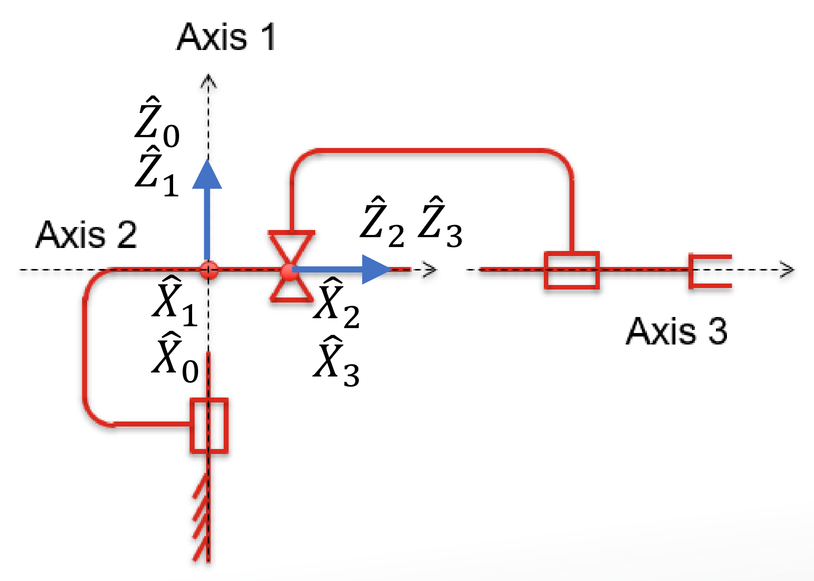
Example 2 – Step 5

- Attach frame $\{n\}$:
  - $\hat{Z}_n$ pointing along the n-th joint axis.
  - $\hat{X}_n$ pointing in the direction such that if the n-th joint parameter (in this example θ_3) is zero, then $\hat{X}_n$ is aligned with $\hat{X}_{n-1}$.



Example 2 – Step 6

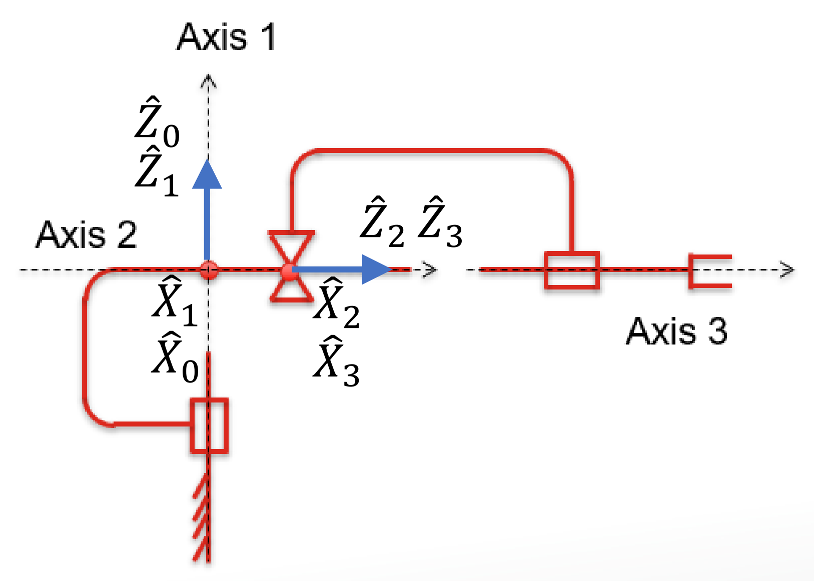
- Put in the link lengths a_0 to a_{n-1}.
- Definition: a_{i-1} = distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ along $\hat{X}_{i-1}$.
  - a_0 = distance from $\hat{Z}_0$ to $\hat{Z}_1$ along $\hat{X}_0$ = 0.
  - a_1 = distance from $\hat{Z}_1$ to $\hat{Z}_2$ along $\hat{X}_1$ = 0 (intercept).
  - a_2 = distance from $\hat{Z}_2$ to $\hat{Z}_3$ along $\hat{X}_2$ = 0 (intercept).



Example 2 – Step 7

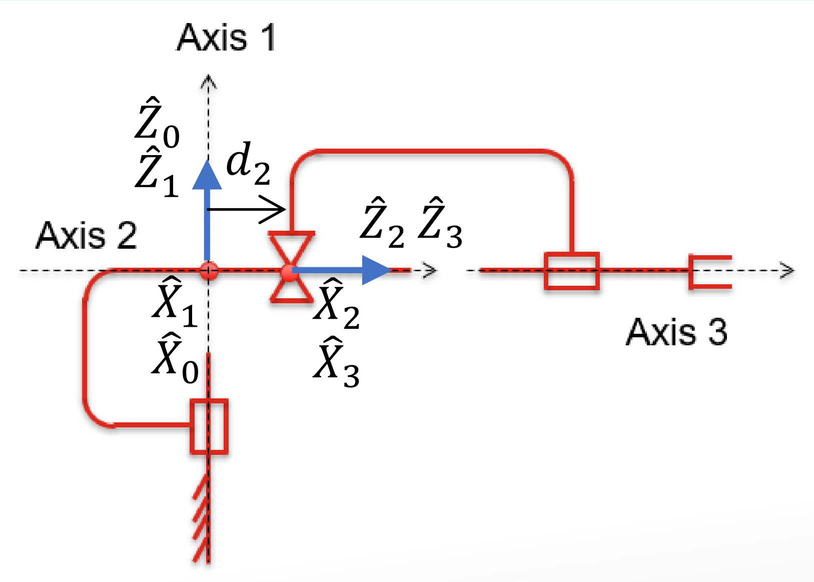
- Put in the link twists α_0 to α_{n-1}.

- Definition: α_{i-1} = angle between $\hat{Z}_{i-1}$ to $\hat{Z}_i$ about $\hat{X}_{i-1}$.
  - α_0 = angle between $\hat{Z}_0$ to $\hat{Z}_1$ about $\hat{X}_0$ = 0.
  - α_1 = angle between $\hat{Z}_1$ to $\hat{Z}_2$ about $\hat{X}_1$ = -90.
  - α_2 = angle between $\hat{Z}_2$ to $\hat{Z}_3$ about $\hat{X}_2$ = 0.



Example 2 – Step 8

- Put in the link offsets d_1 to d_n.
- Definition: d_i = distance from $\hat{X}_{i-1}$ to $\hat{X}_i$ along $\hat{Z}_i$.
  - d_1 = distance from $\hat{X}_0$ to $\hat{X}_1$ along $\hat{Z}_1$ = 0.
  - d_2 = distance from $\hat{X}_1$ to $\hat{X}_2$ along $\hat{Z}_2$ = variable.
  - d_3 = distance from $\hat{X}_2$ to $\hat{X}_3$ along $\hat{Z}_3$ = 0.

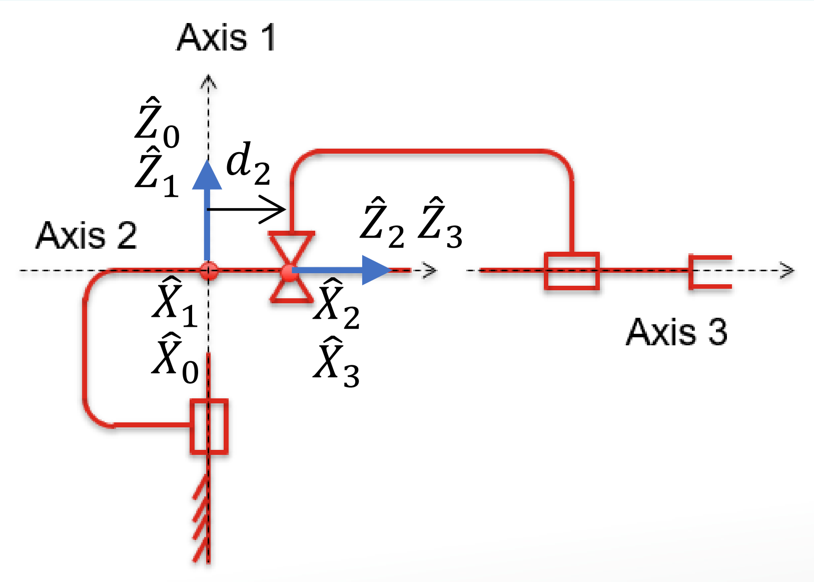


Example 2 – Step 9

- Put in the joint angles θ_1 to θ_n.

- Definition: θ_i = angle btw. $\hat{X}_{i-1}$ to $\hat{X}_i$ about $\hat{Z}_i$.

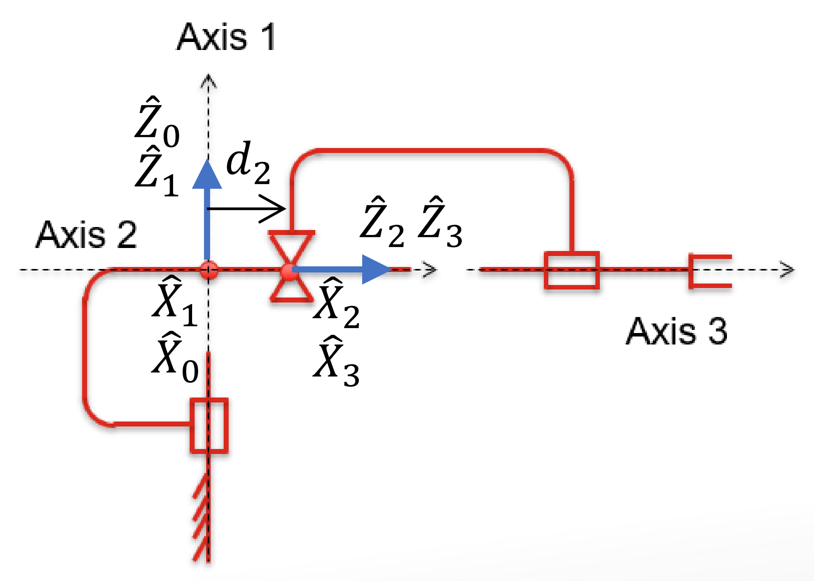
  - θ_1 = angle btw. $\hat{X}_0$ to $\hat{X}_1$ about $\hat{Z}_1$ = variable.
  - θ_2 = angle btw. $\hat{X}_1$ to $\hat{X}_2$ about $\hat{Z}_2$ = 0.
  - θ_3 = angle btw. $\hat{X}_2$ to $\hat{X}_3$ about $\hat{Z}_3$ = variable.



Example 2 – Step 10

- Transfer the results into a DH table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	0
3	0	0	0	θ_3



Example 2 – Step 11

- Calculate the transformations.

$$
{}^{i-1}_{i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

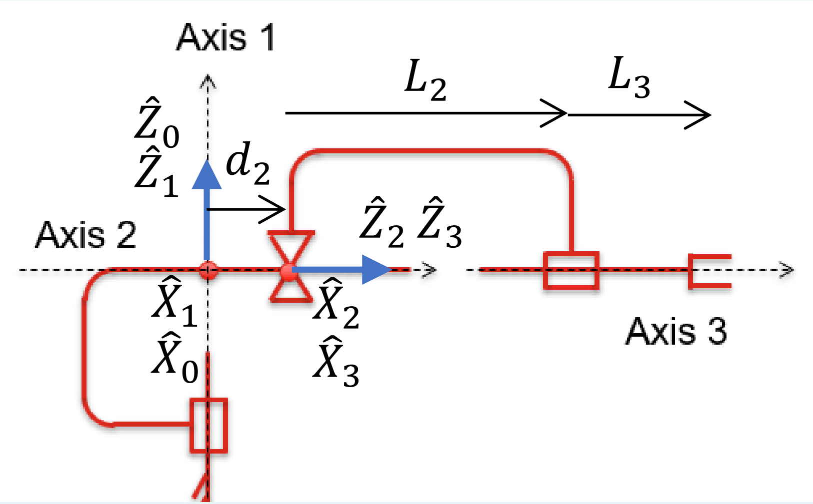
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	0
3	0	0	0	θ_3

$$
{}^{0}_{1}T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; {}^{1}_{2}T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; {}^{2}_{3}T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

$$
{}^{0}_{3}T = {}^{0}_{1}T \cdot {}^{1}_{2}T \cdot {}^{2}_{3}T = \begin{bmatrix} c\theta_1 c\theta_3 & -c\theta_1 s\theta_3 & -s\theta_1 & -d_2 s\theta_1 \\ s\theta_1 c\theta_3 & -s\theta_1 s\theta_3 & c\theta_1 & d_2 c\theta_1 \\ -s\theta_3 & -c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Example 2 – Step 12

- By now, we have already obtained the position and orientation of frame {3}. What about the end-effector?
- According to the figure, it is at ${}^{3}P = [0 \quad 0 \quad L_2 + L_3]^T$. This is a constant! The end-effector doesn't move wrt. frame {3}!



Example 2 – Step 13

- Calculate the position and orientation of end-effector wrt. frame {0}.

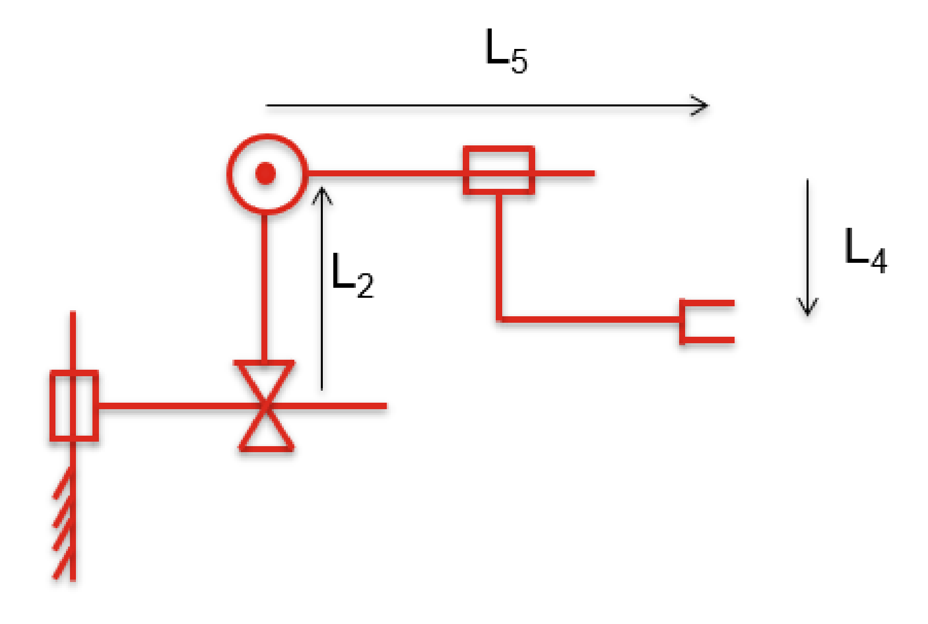
$$
{}^{0}P = {}^{0}_{3}T \cdot {}^{3}P = \begin{bmatrix} c\theta_1 c\theta_3 & -c\theta_1 s\theta_3 & -s\theta_1 & -d_2 s\theta_1 \\ s\theta_1 c\theta_3 & -s\theta_1 s\theta_3 & c\theta_1 & d_2 c\theta_1 \\ -s\theta_3 & -c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ L_2 + L_3 \\ 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} -(L_2 + L_3 + d_2) s\theta_1 \\ (L_2 + L_3 + d_2) c\theta_1 \\ 0 \\ 1 \end{bmatrix}
$$

- Sanity check, e.g. when $\theta_1 = 0$. Result makes sense!

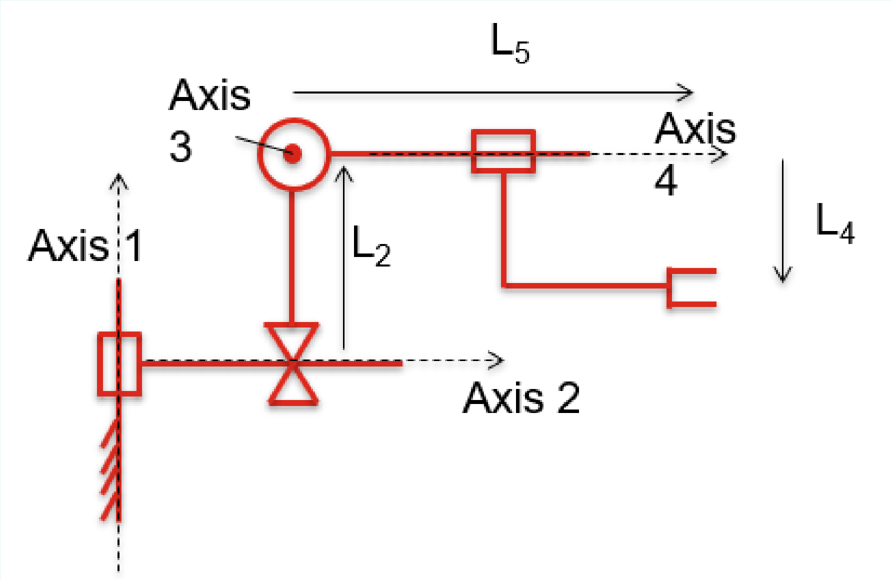
Example 3

- 4-link RPRR non-planar robot :



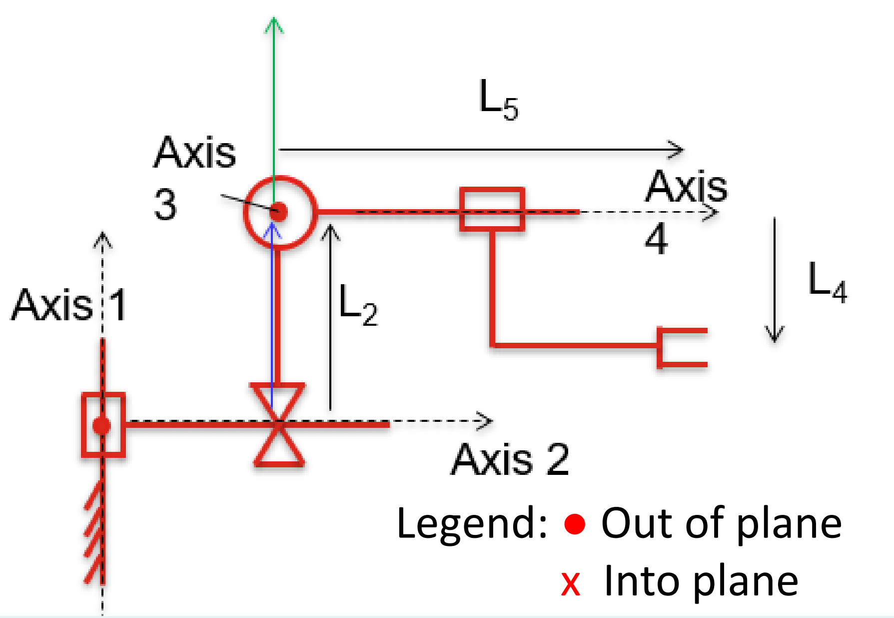
Example 3 – Step 1

- Draw axes:
  - For revolute joints: About the rotation.
  - For prismatic joints: Along the translation.
  - The direction of arrow is arbitrary, but once fixed must be followed through til the end.



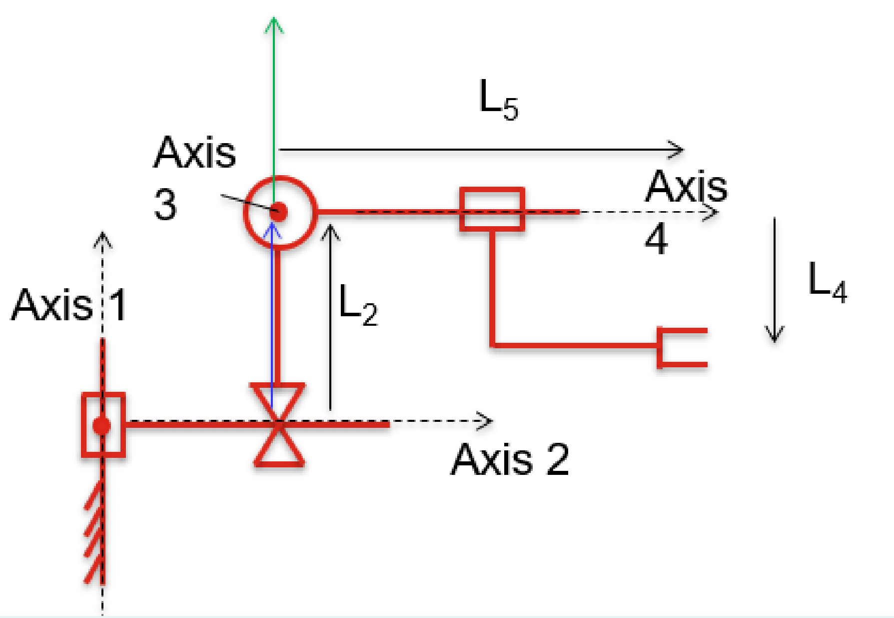
Example 3 – Step 2

- Draw the mutual perpen-dicular lines between axes.
  - Draw from lower to higher axis, e.g. 1 to 2, 2 to 3…
  - In this example, Axis 1 and Axis 2 intersect.
    - Line is perpendicular to the plane made by axes 1 and 2.
    - Because the axes intersect, the sense of direction is arbitrary (either into or out of the plane). Here I have chosen out of the plane.



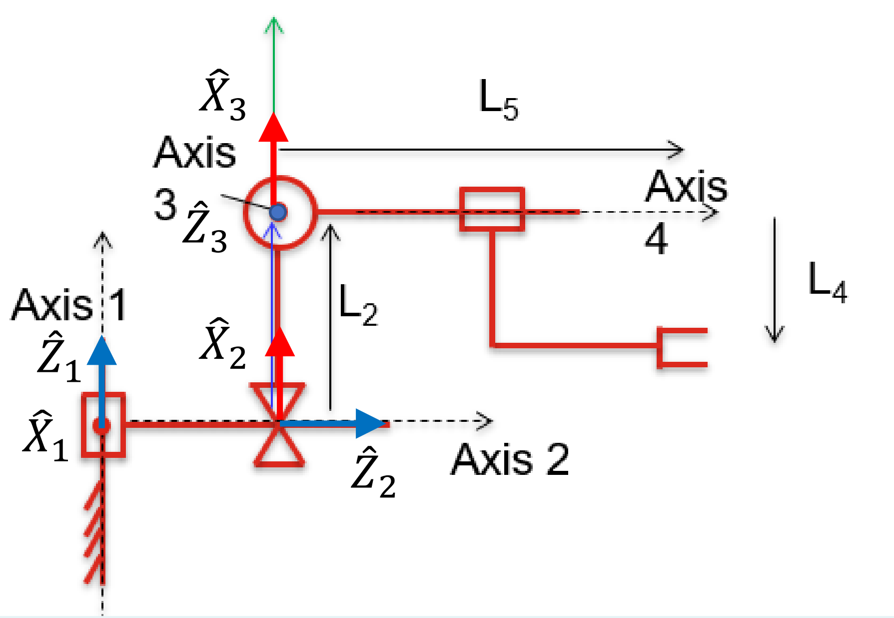
Example 3 – Step 2 Continued

- Between Axis 2 and Axis 3
  - Blue arrow as shown.
- Axis 3 and Axis 4 intersect.
  - Line is perpendicular to the plane made by axes 3 and 4.
  - Because the axes intersect, the sense of direction is arbitrary (either up or down).
  - Here I have chosen up. Green arrow as shown.



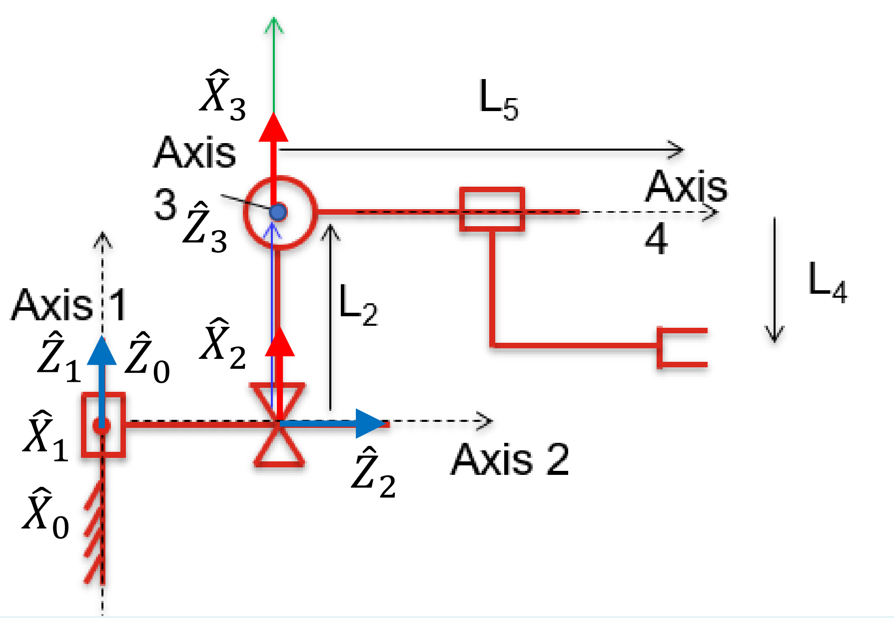
Example 3 – Step 3

- Attach frames {1} to $\{n-1\}$ as follows:
  - $\hat{Z}_i$ pointing along the i-th joint axis.
  - $\hat{X}_i$ pointing along mutual perpendicular.
  - (Assign $\hat{Y}_i$ to complete the right-hand coordinate system.)



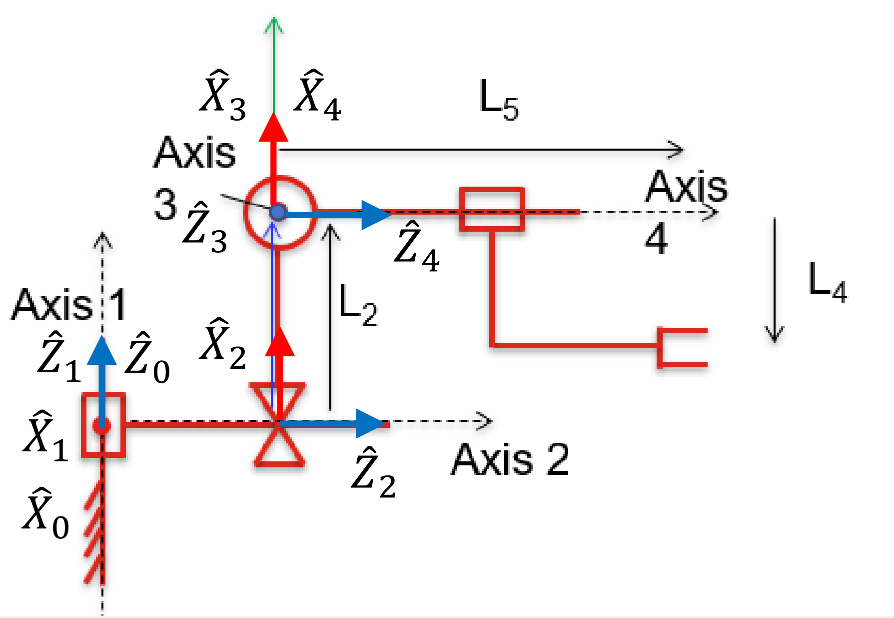
Example 3 – Step 4

- Attach frame {0}:
  - Assign {0} to match {1} WHEN the first joint variable (in this case θ_1) is zero.



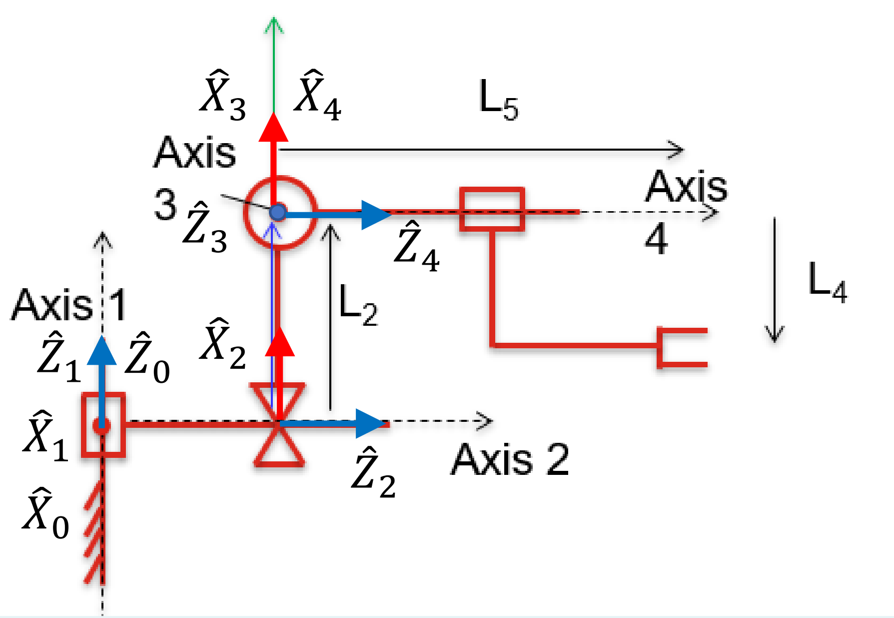
Example 3 – Step 5

- Attach frame $\{n\}$:
  - $\hat{Z}_n$ pointing along the n-th joint axis.
  - $\hat{X}_n$ pointing in the direction such that if the n-th joint parameter (in this example θ_4) is zero, then $\hat{X}_n$ is aligned with $\hat{X}_{n-1}$.



Example 3 – Step 6

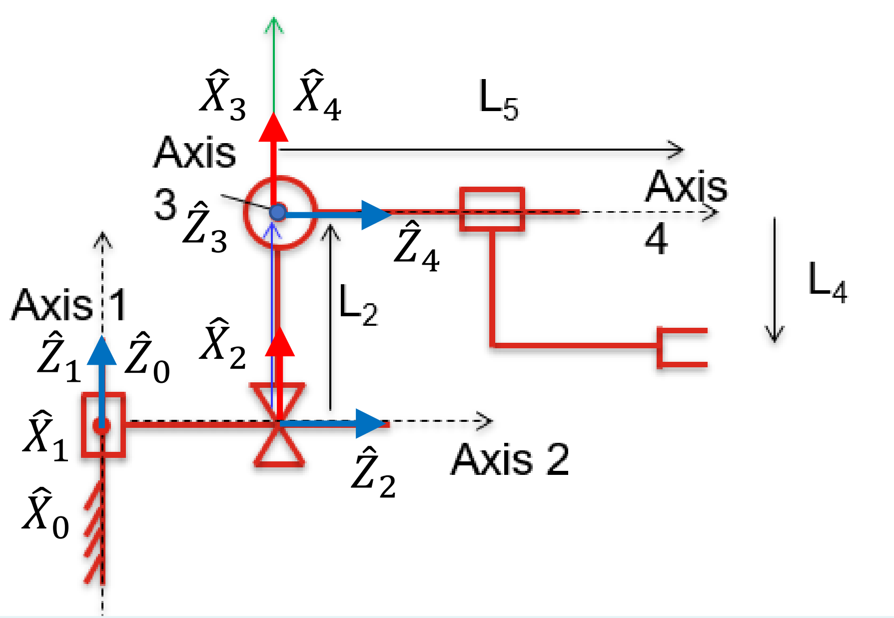
- Put in link lengths a_0 to a_{n-1}.
- Definition: a_{i-1} = distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ along $\hat{X}_{i-1}$.
  - a_0 = distance from $\hat{Z}_0$ to $\hat{Z}_1$ along $\hat{X}_0$ = 0.
  - a_1 = distance from $\hat{Z}_1$ to $\hat{Z}_2$ along $\hat{X}_1$ = 0 (intercept).
  - a_2 = distance from $\hat{Z}_2$ to $\hat{Z}_3$ along $\hat{X}_2 = L_2$.
  - a_3 = distance from $\hat{Z}_3$ to $\hat{Z}_4$ along $\hat{X}_3$ = 0 (intercept).



Example 3 – Step 7

- Put in link twists α_0 to α_{n-1}.

- Definition: α_{i-1} = angle between $\hat{Z}_{i-1}$ to $\hat{Z}_i$ about $\hat{X}_{i-1}$.
  - α_0 = angle between $\hat{Z}_0$ to $\hat{Z}_1$ about $\hat{X}_0$ = 0.
  - α_1 = angle between $\hat{Z}_1$ to $\hat{Z}_2$ about $\hat{X}_1$ = -90.
  - α_2 = angle between $\hat{Z}_2$ to $\hat{Z}_3$ about $\hat{X}_2$ = -90.
  - α_3 = angle between $\hat{Z}_3$ to $\hat{Z}_4$ about $\hat{X}_3$ = 90.

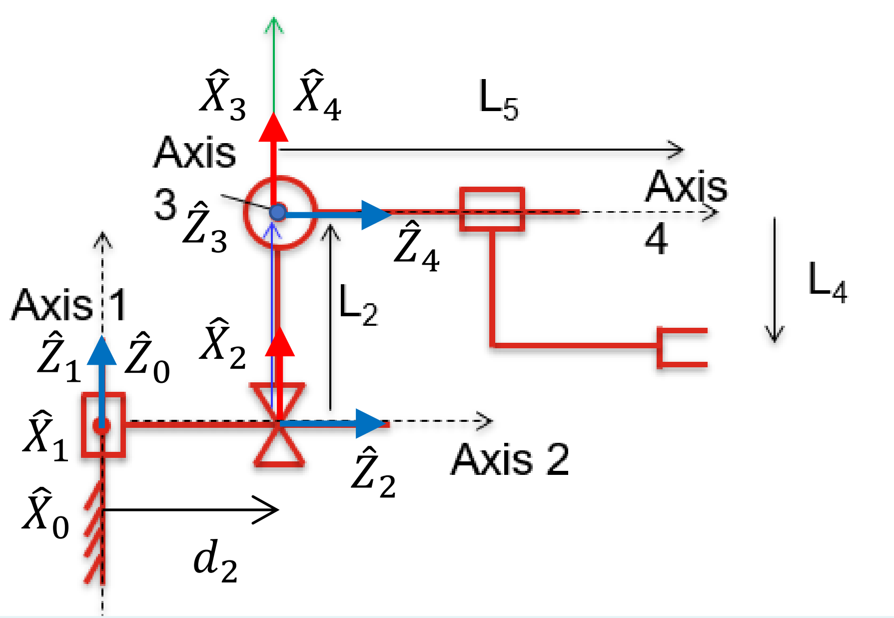


Example 3 – Step 8

- Put in link offsets d_1 to d_n.

- Definition: d_i = distance from $\hat{X}_{i-1}$ to $\hat{X}_i$ along $\hat{Z}_i$.

  - d_1 = distance from $\hat{X}_0$ to $\hat{X}_1$ along $\hat{Z}_1$ = 0.
  - d_2 = distance from $\hat{X}_1$ to $\hat{X}_2$ along $\hat{Z}_2$ = variable.
  - d_3 = distance from $\hat{X}_2$ to $\hat{X}_3$ along $\hat{Z}_3$ = 0.
  - d_4 = distance from $\hat{X}_3$ to $\hat{X}_4$ along $\hat{Z}_4$ = 0.

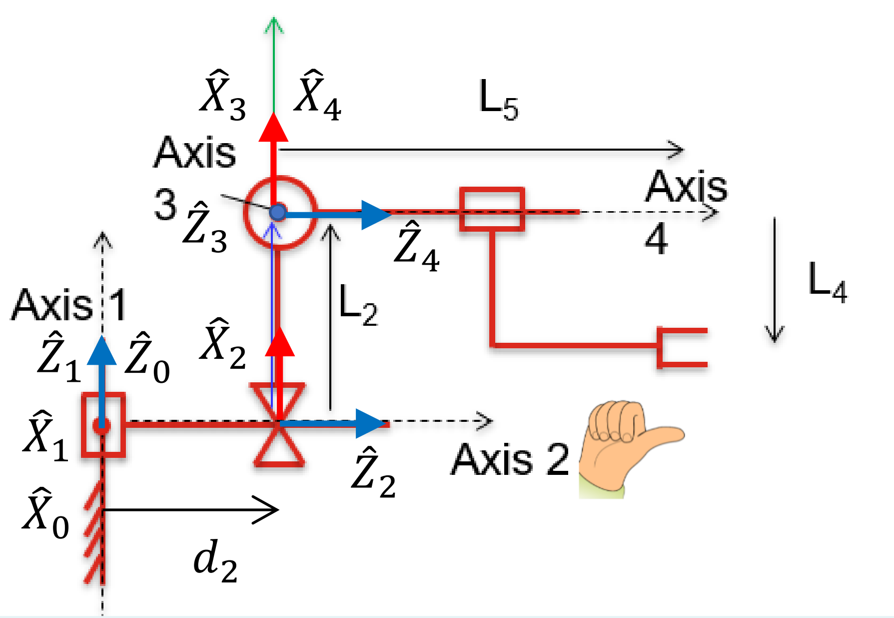


Example 3 – Step 9

- Put in joint angles θ_1 to θ_n.

- Definition: θ_i = angle btw. $\hat{X}_{i-1}$ to $\hat{X}_i$ about $\hat{Z}_i$.

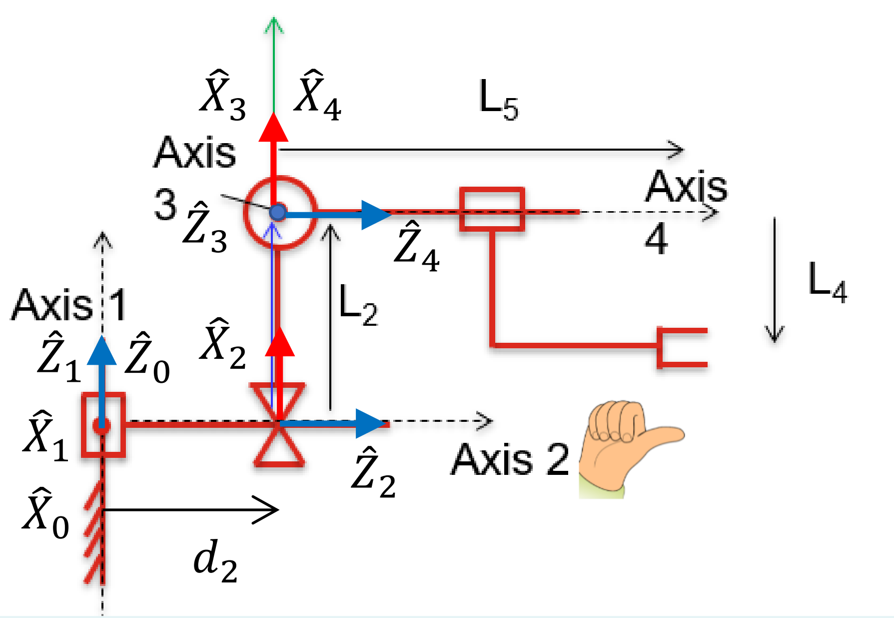
  - θ_1 = angle btw. $\hat{X}_0$ to $\hat{X}_1$ about $\hat{Z}_1$ = variable.
  - θ_2 = angle btw. $\hat{X}_1$ to $\hat{X}_2$ about $\hat{Z}_2$ = -90.
  - θ_3 = angle btw. $\hat{X}_2$ to $\hat{X}_3$ about $\hat{Z}_3$ = variable.
  - θ_4 = angle btw. $\hat{X}_3$ to $\hat{X}_4$ about $\hat{Z}_4$ = variable.



Example 3 – Step 10

- Transfer the results into a DH table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	-90
3	-90	L_2	0	θ_3
4	90	0	0	θ_4



Example 3 – Step 11

- Calculate the transformations.

$$ {}^{i-1}_{i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

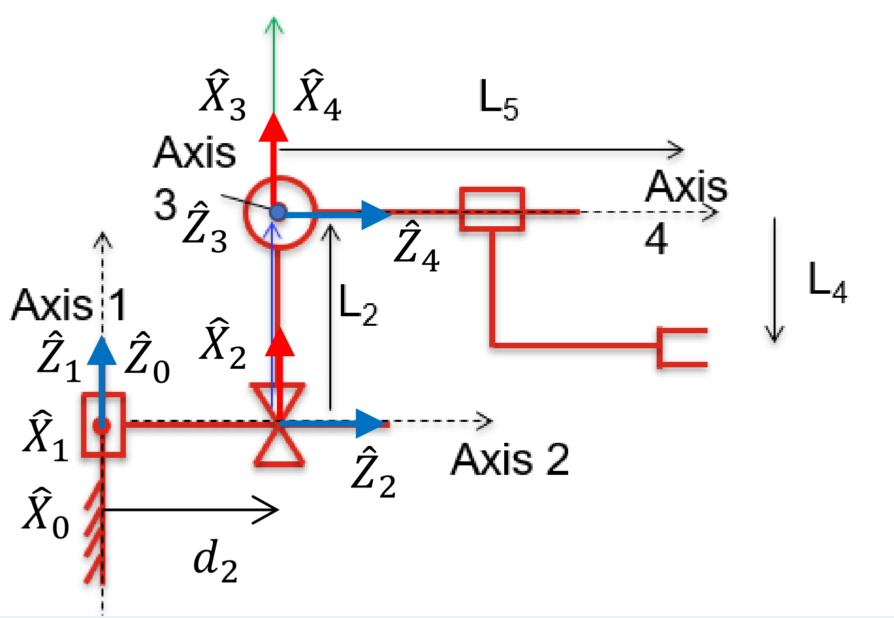
i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	-90
3	-90	L_2	0	θ_3
4	90	0	0	θ_4

$$ {}^{0}_{1}T = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{1}_{2}T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{2}_{3}T = \begin{bmatrix} c3 & -s3 & 0 & L_2 \\ 0 & 0 & 1 & 0 \\ -s3 & -c3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{3}_{4}T = \begin{bmatrix} c4 & -s4 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s4 & c4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{0}_{4}T = {}^{0}_{1}T \cdot {}^{1}_{2}T \cdot {}^{2}_{3}T \cdot {}^{3}_{4}T = \begin{bmatrix} s1s3s4 + c1s4 & -s1s3s4 + c1c4 & -s1c3 & -d_2 s1 \\ -c1s3c4 + s1s4 & c1s3s4 + s1c4 & c1c3 & d_2 c1 \\ c3c4 & -c3s4 & s3 & L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Example 3 – Step 12

- By now, we have already obtained the position and orientation of frame {4}. What about the end-effector?
- Based on the figure, it is at ${}^4P = [-L_4 \quad 0 \quad L_5]^T$. This is a constant! The end-effector doesn't move wrt. frame {4}!



Example 3 – Step 13

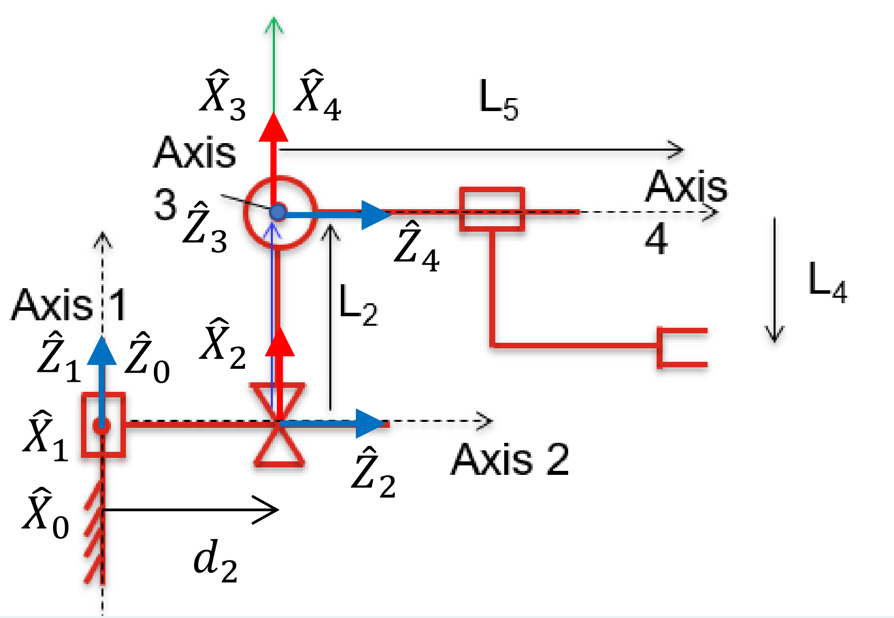
- Calculate the position and orientation of end-effector wrt. frame {0}.

$$
{}^{0}P = {}^{0}_{4}T \cdot {}^{4}P = \begin{bmatrix} s1s3s4 + c1s4 & -s1s3s4 + c1c4 & -s1c3 & -d_2s1 \\ -c1s3c4 + s1s4 & c1s3s4 + s1c4 & c1c3 & d_2c1 \\ c3c4 & -c3s4 & s3 & L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -L_4 \\ 0 \\ L_5 \\ 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} -L_4(s1s3s4 + c1s4) - L_5s1c3 - d_2s1 \\ -L_4(-c1s3c4 + s1s4) + L_5c1c3 + d_2c1 \\ -L_4c3c4 + L_5s3 + L_2 \\ 1 \end{bmatrix}
$$

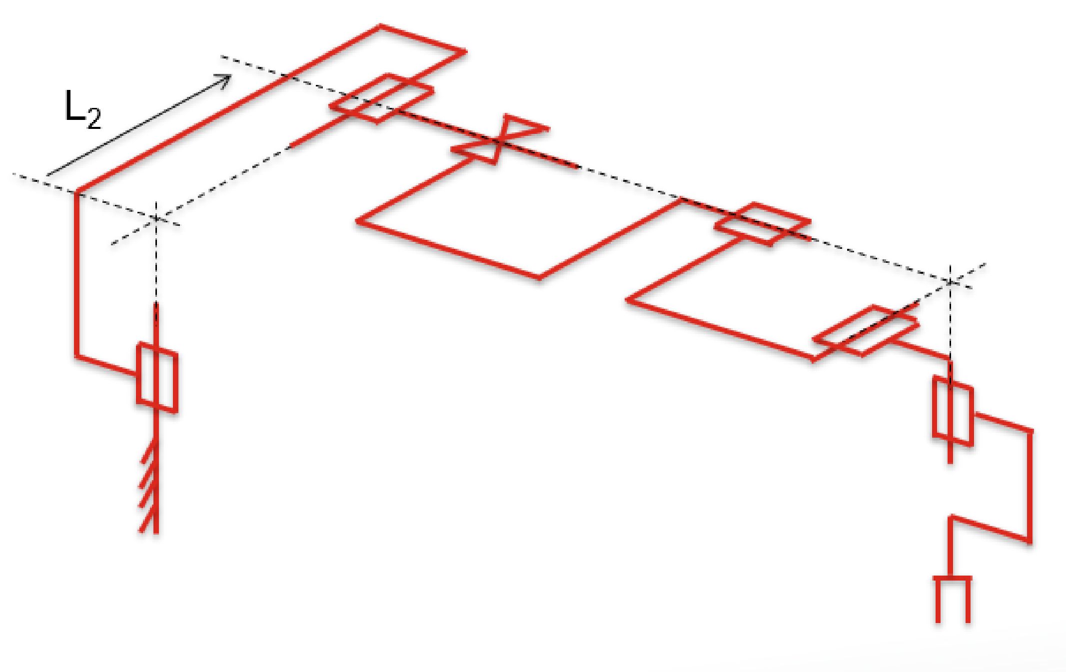
Example 3 – Step 13 Continued

- Sanity check:
- In the configuration as shown:
  - $\theta_1 = 0$
  - $\theta_2 = -90$ (remember: constant)
  - $\theta_3 = 0$
  - $\theta_4 = 0$
  - (See definition for θ!)
- Using these values gives:
  - $^{o}P = \begin{bmatrix} 0 \\ L_5 + d_2 \\ -L_4 + L_2 \\ 1 \end{bmatrix}$
  - Result makes sense!



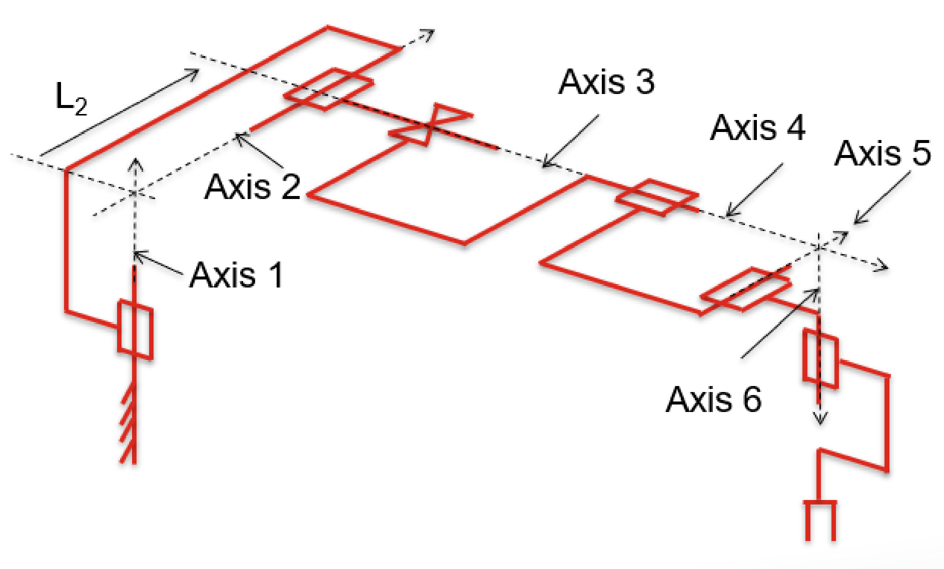
Example 4

- 6-link Stanford Scheinman robot:



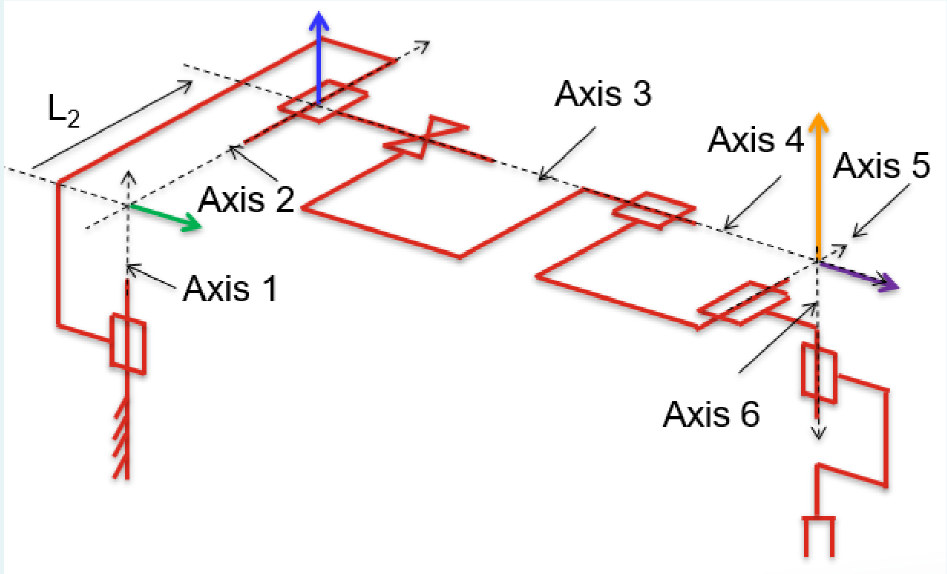
Example 4 – Step 1

- Draw axes:
  - For revolute joints: About the rotation.
  - For prismatic joints: Along the translation.
  - The direction of arrow is arbitrary, but once fixed must be followed through til the end.



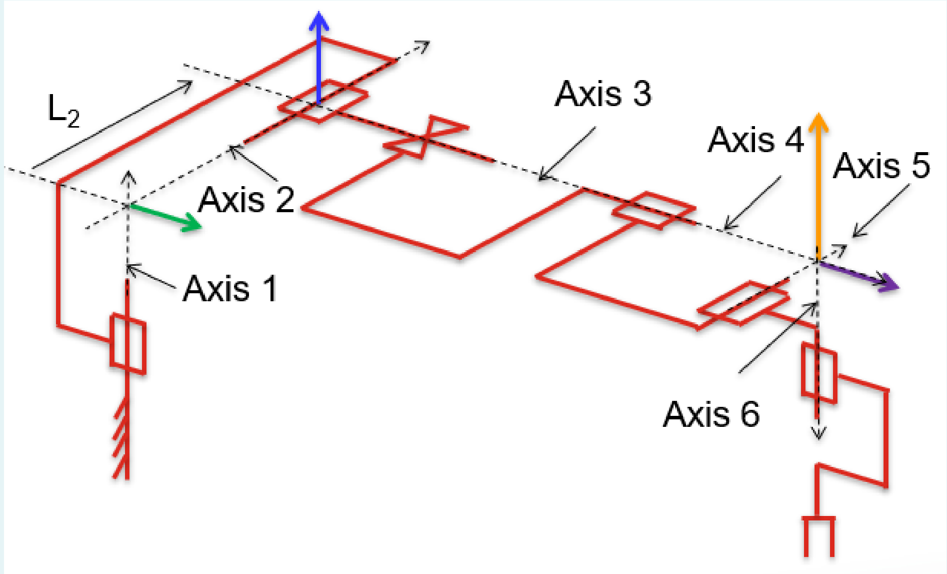
Example 4 – Step 2

- Draw the mutual perpen-dicular lines between axes.
  - Draw from lower to higher axis, e.g. 1 to 2, 2 to 3…
  - In this example:
    - Axis 1 and Axis 2 intersect → Green arrow.
    - Axis 2 and Axis 3 intersect → Blue arrow.



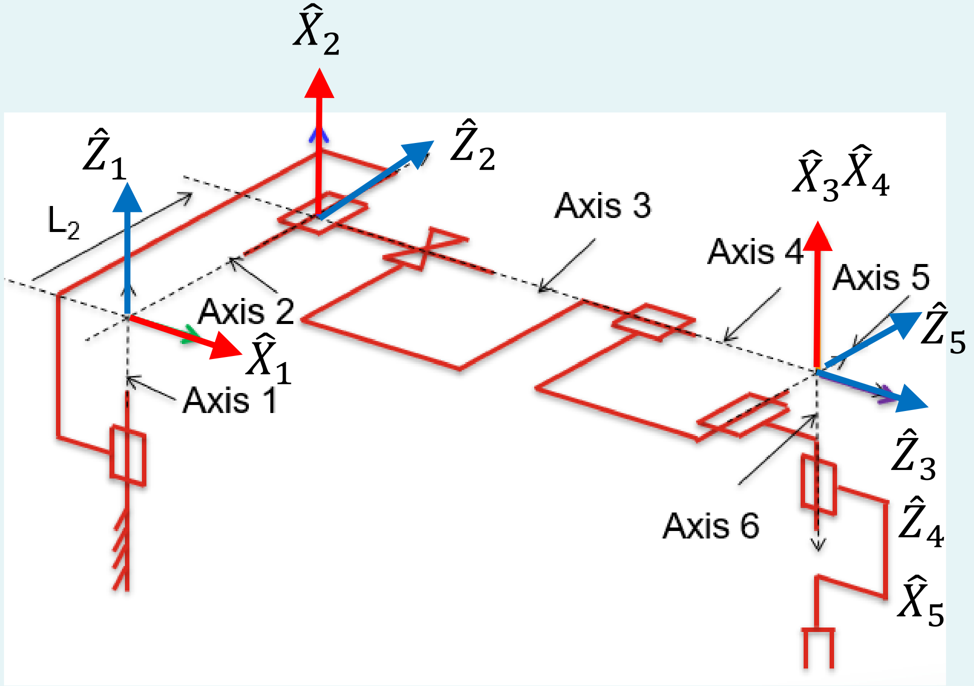
Example 4 – Step 2 Continued

- Axis 3 and Axis 4 are the same → Arbitrary, Orange arrow (same as Axis 4 and 5 below).
- Axis 4 and Axis 5 intersect → Orange arrow.
- Axis 5 and Axis 6 intersect → Purple arrow.



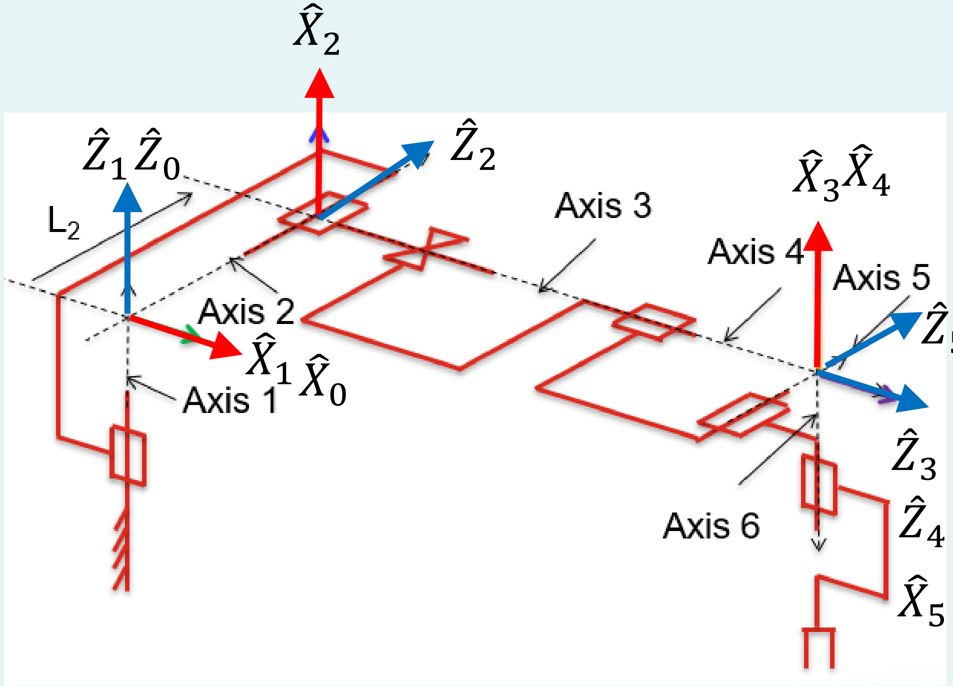
Example 4 – Step 3

- Attach frames {1} to $\{n-1\}$ as follows:
  - $\hat{Z}_i$ pointing along the i-th joint axis.
  - $\hat{X}_i$ pointing along mutual perpendicular.
  - (Assign $\hat{Y}_i$ to complete the right-hand coordinate system.)



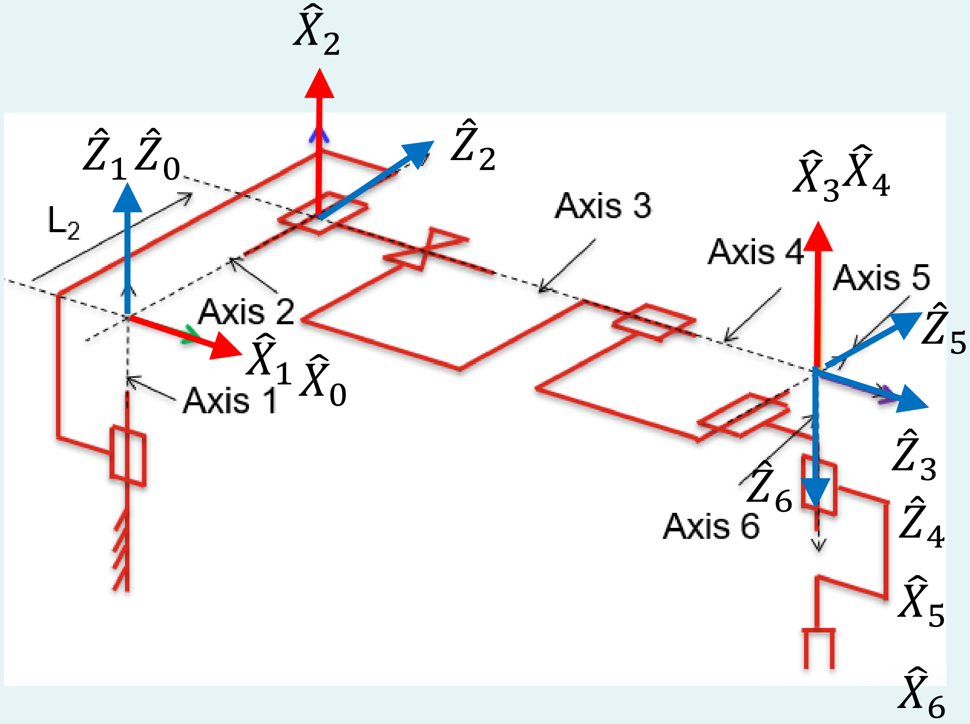
Example 4 – Step 4

- Attach frame {0}:
  - Assign {0} to match {1} WHEN the first joint variable (in this case θ_1) is zero.



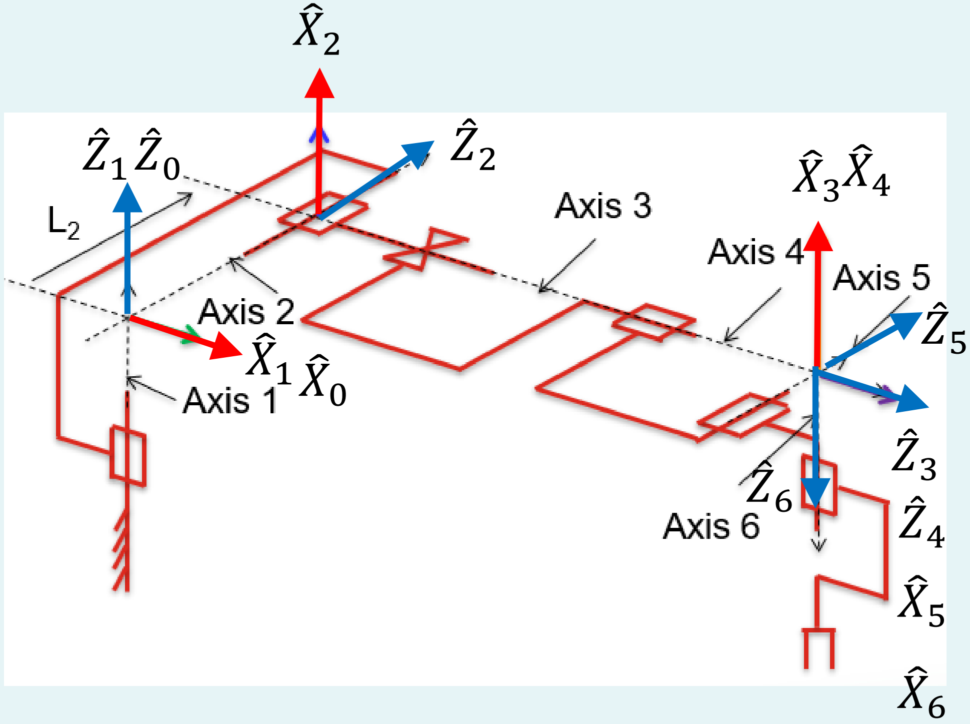
Example 4 – Step 5

- Attach frame $\{n\}$:
  - $\hat{Z}_n$ pointing along the n-th joint axis.
  - $\hat{X}_n$ pointing in the direction such that if the n-th joint parameter (in this example θ_6) is zero, then $\hat{X}_n$ is aligned with $\hat{X}_{n-1}$.



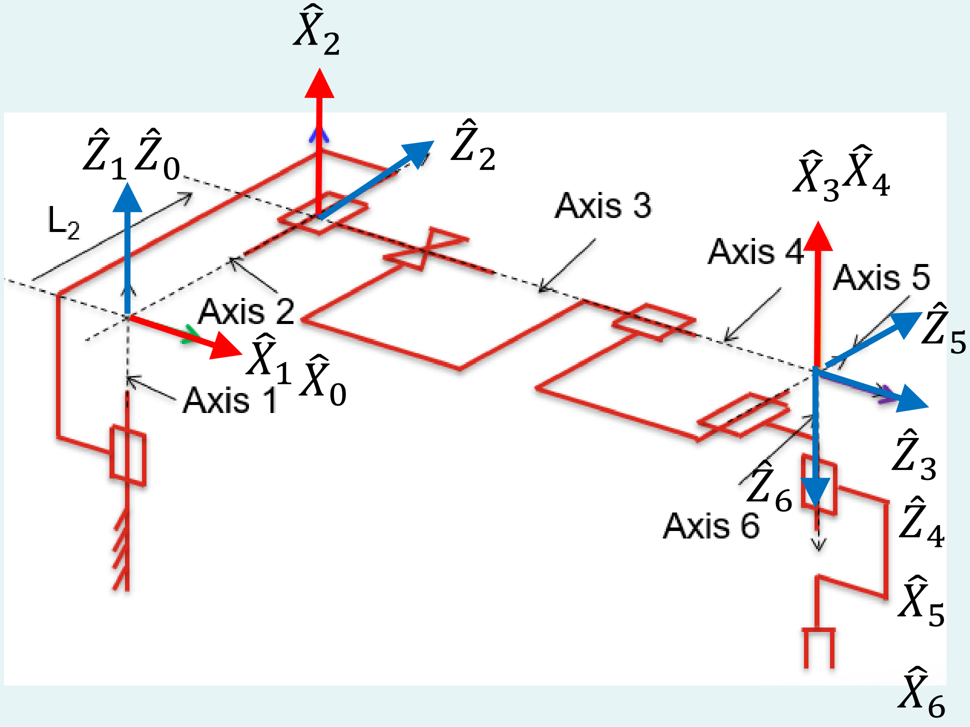
Example 4 – Step 6

- Put in link lengths a_0 to a_{n-1}.
- Definition: a_{i-1} = distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ along $\hat{X}_{i-1}$.
  - a_0 = distance from $\hat{Z}_0$ to $\hat{Z}_1$ along $\hat{X}_0$ = 0.
  - a_1 = distance from $\hat{Z}_1$ to $\hat{Z}_2$ along $\hat{X}_1$ = 0 (intercept).
  - a_2 = distance from $\hat{Z}_2$ to $\hat{Z}_3$ along $\hat{X}_2$ = 0 (intercept).



Example 4 – Step 6 Continued

- a_3 = distance from $\hat{Z}_3$ to $\hat{Z}_4$ along $\hat{X}_3$ = 0.
- a_4 = distance from $\hat{Z}_4$ to $\hat{Z}_5$ along $\hat{X}_4$ = 0 (intercept).
- a_5 = distance from $\hat{Z}_5$ to $\hat{Z}_6$ along $\hat{X}_5$ = 0 (intercept).

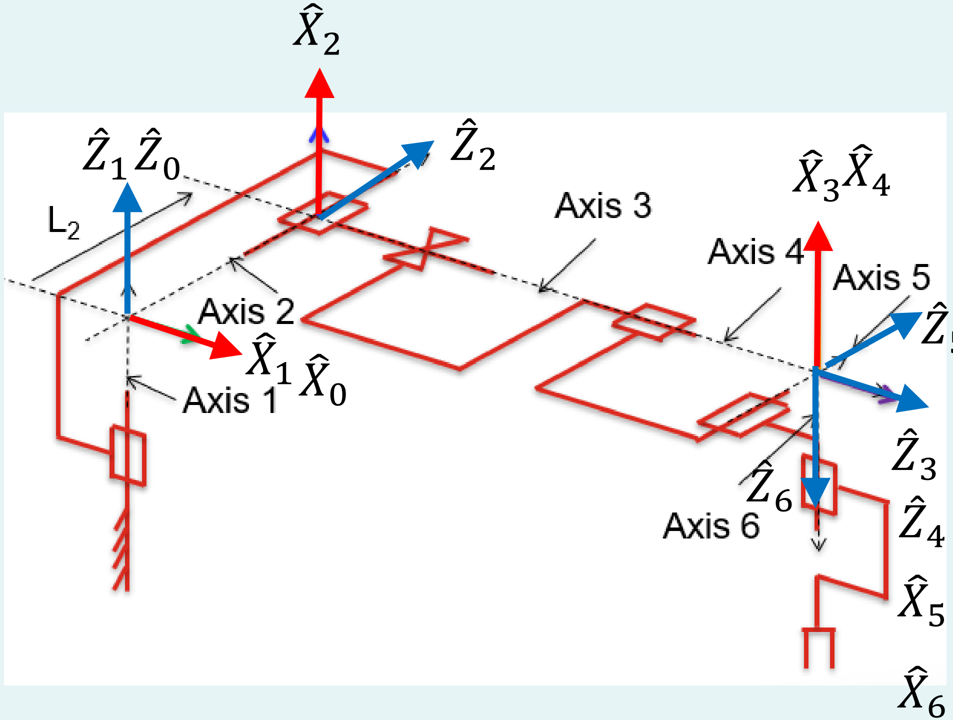


Example 4 – Step 7

- Put in link twists α_0 to α_{n-1}.

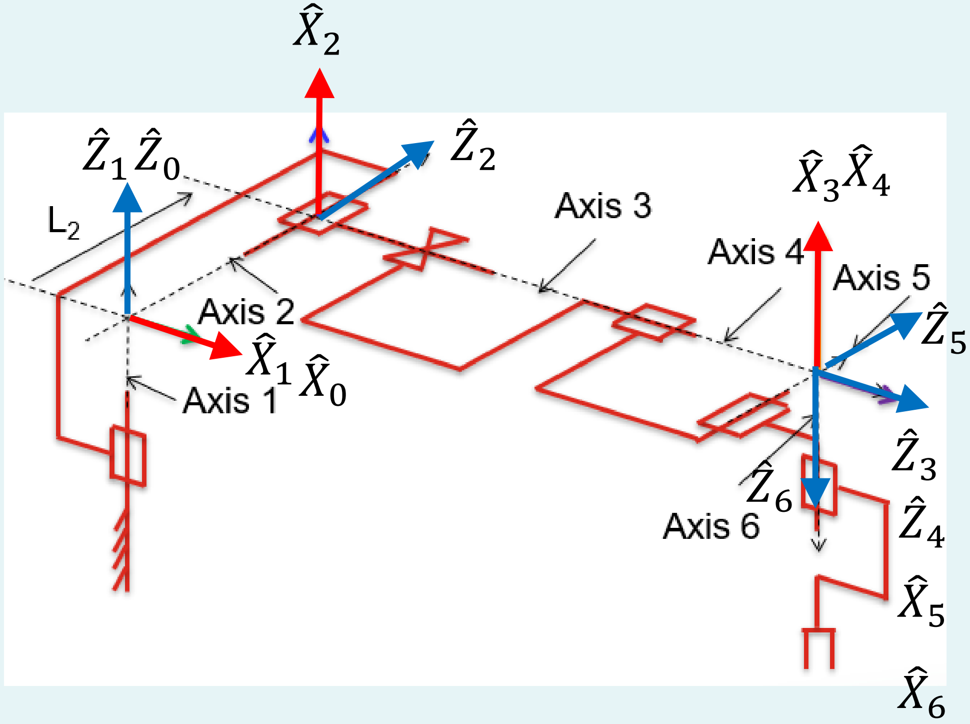
- Definition: α_{i-1} = angle between $\hat{Z}_{i-1}$ to $\hat{Z}_i$ about $\hat{X}_{i-1}$.

  - α_0 = angle between $\hat{Z}_0$ to $\hat{Z}_1$ about $\hat{X}_0$ = 0.
  - α_1 = angle between $\hat{Z}_1$ to $\hat{Z}_2$ about $\hat{X}_1$ = -90.
  - α_2 = angle between $\hat{Z}_2$ to $\hat{Z}_3$ about $\hat{X}_2$ = -90.



Example 4 – Step 7 Continued

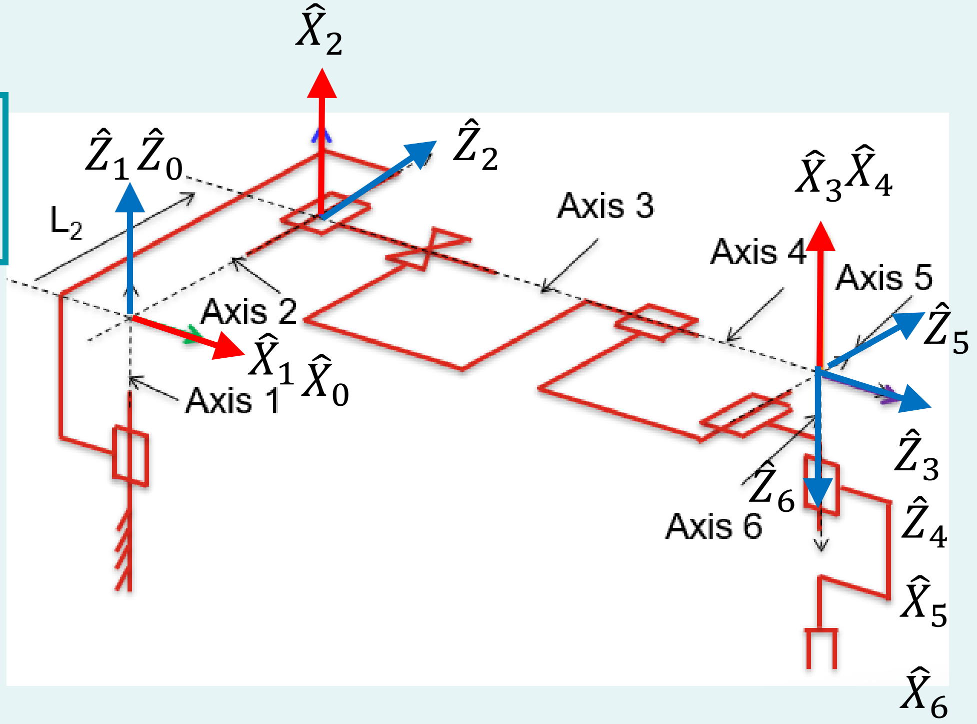
- α_3 = angle between $\hat{Z}_3$ to $\hat{Z}_4$ about $\hat{X}_3$ = 0.
- α_4 = angle between $\hat{Z}_4$ to $\hat{Z}_5$ about $\hat{X}_4$ = 90.
- α_5 = angle between $\hat{Z}_5$ to $\hat{Z}_6$ about $\hat{X}_5$ = -90.



Example 4 – Step 8

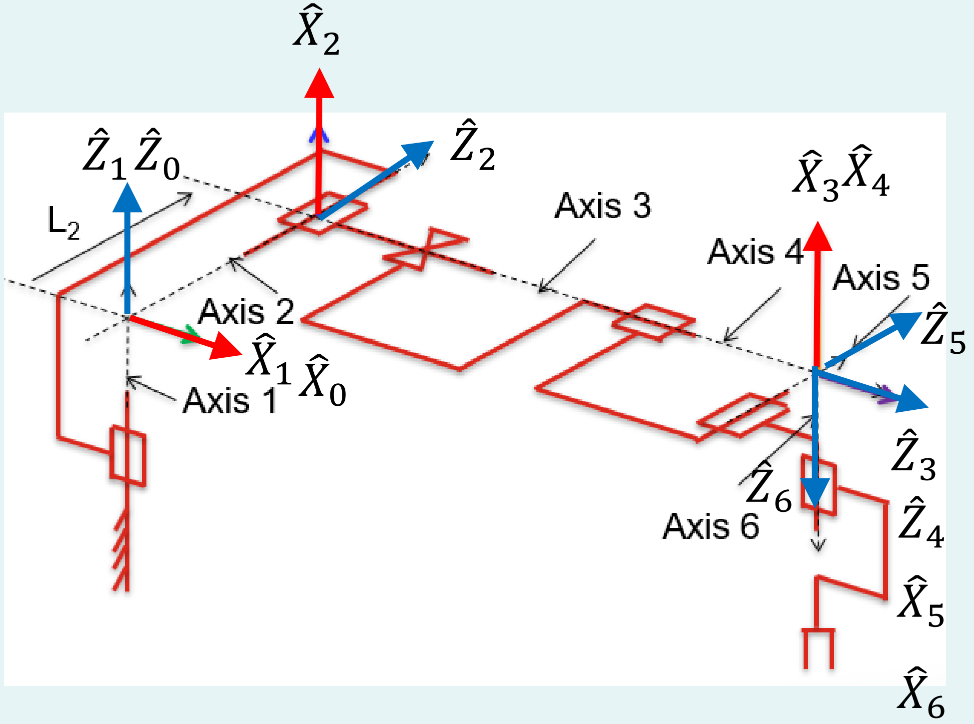
- Put in link offsets d_1 to d_n.

- Definition: d_i = distance from $\hat{X}_{i-1}$ to $\hat{X}_i$ along $\hat{Z}_i$.
    - d_1 = distance from $\hat{X}_0$ to $\hat{X}_1$ along $\hat{Z}_1$ = 0.
    - d_2 = distance from $\hat{X}_1$ to $\hat{X}_2$ along $\hat{Z}_2 = L_2$.
    - d_3 = distance from $\hat{X}_2$ to $\hat{X}_3$ along $\hat{Z}_3$ = variable.



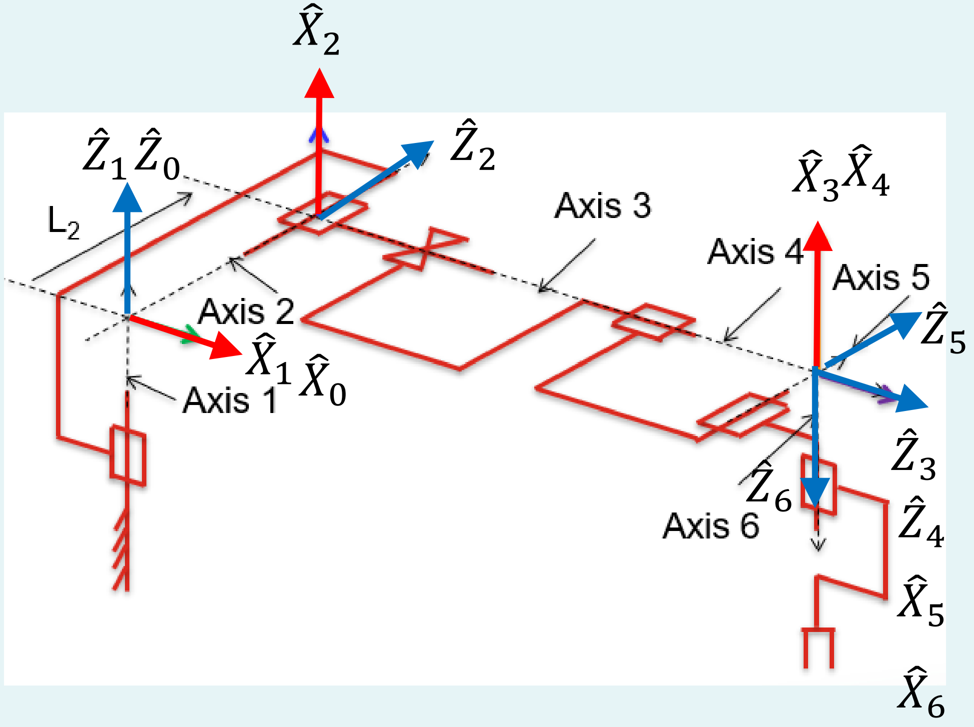
Example 4 – Step 8 Continued

- d_4 = distance from $\hat{X}_3$ to $\hat{X}_4$ along $\hat{Z}_4$ = 0.
- d_5 = distance from $\hat{X}_4$ to $\hat{X}_5$ along $\hat{Z}_5$ = 0.
- d_6 = distance from $\hat{X}_5$ to $\hat{X}_6$ along $\hat{Z}_6$ = 0.



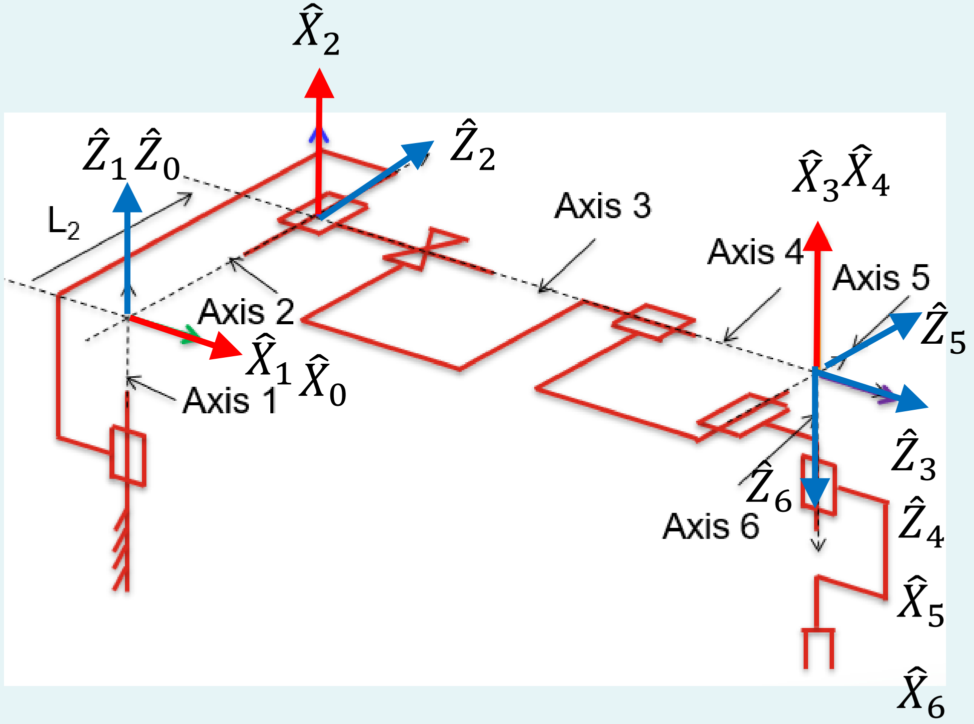
Example 4 – Step 9

- Put in joint angles θ_1 to θ_n.
- Definition: θ_i = angle btw. $\hat{X}_{i-1}$ to $\hat{X}_i$ about $\hat{Z}_i$.
  - θ_1 = angle btw. $\hat{X}_0$ to $\hat{X}_1$ about $\hat{Z}_1$ = variable.
  - θ_2 = angle btw. $\hat{X}_1$ to $\hat{X}_2$ about $\hat{Z}_2$ = variable.
  - θ_3 = angle btw. $\hat{X}_2$ to $\hat{X}_3$ about $\hat{Z}_3$ = 0.



Example 4 – Step 9 Continued

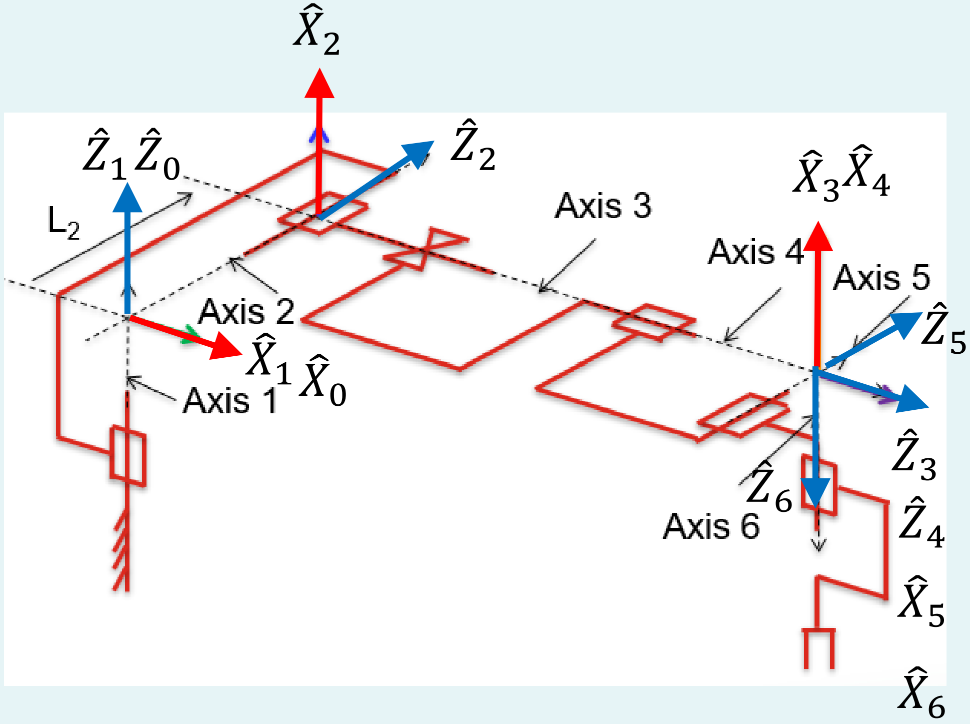
- θ_4 = angle btw. $\hat{X}_3$ to $\hat{X}_4$ about $\hat{Z}_4$ = variable.
- θ_5 = angle btw. $\hat{X}_4$ to $\hat{X}_5$ about $\hat{Z}_5$ = variable.
- θ_6 = angle btw. $\hat{X}_5$ to $\hat{X}_6$ about $\hat{Z}_6$ = variable.



Example 4 – Step 10

- Transfer the results into a DH table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	L_2	θ_2
3	-90	0	d_3	0
4	0	0	0	θ_4
5	90	0	0	θ_5
6	-90	0	0	θ_6



Example 4 – Step 11

- Calculate the transformations.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	L_2	θ_2
3	-90	0	d_3	0
4	0	0	0	θ_4
5	90	0	0	θ_5
6	-90	0	0	θ_6

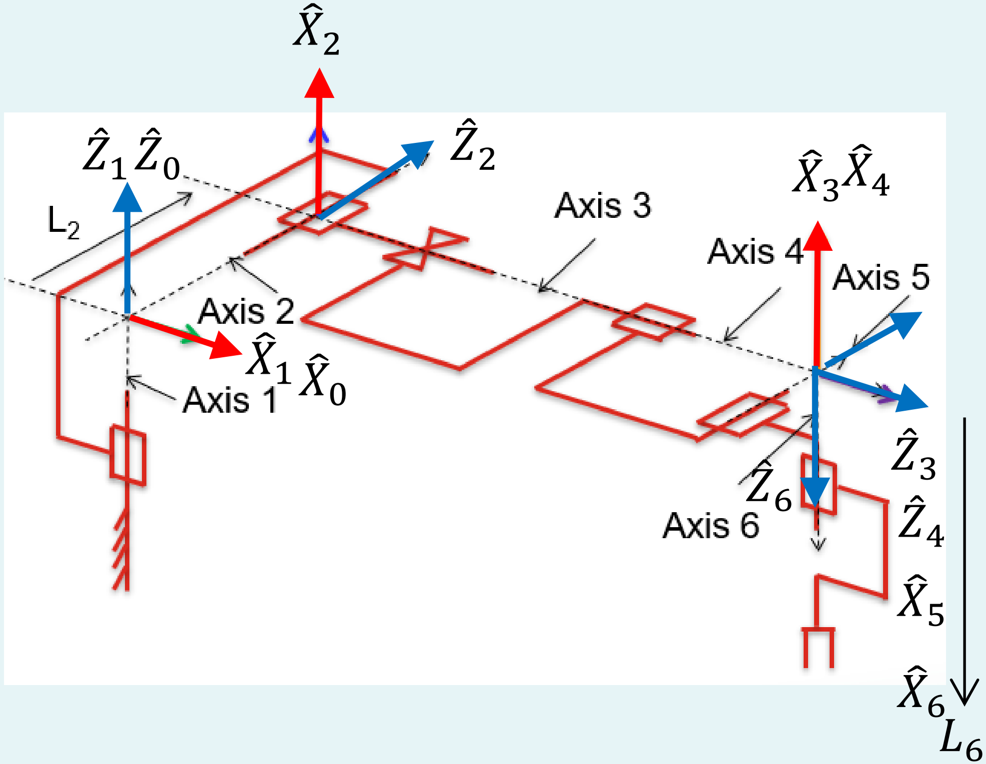
$$\boxed{{}^{i-1}_{\ \ i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

$${}^0_1T = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c2 & -s2 & 0 & 0 \\ 0 & 0 & 1 & L_2 \\ -s2 & -c2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^2_3T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$${}^3_4T = \begin{bmatrix} c4 & -s4 & 0 & 0 \\ s4 & c4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^4_5T = \begin{bmatrix} c5 & -s5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s5 & c5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^5_6T = \begin{bmatrix} c6 & -s6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s6 & -c6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^0_6T = \prod_{i=1}^{6} {}^{i-1}_{\ \ i}T$$

Example 4 – Step 12

- By now, we have already obtained the position and orientation of frame {6}. What about the end-effector?
- Based on the figure, it is at ${}^{6}P = [0 \quad 0 \quad L_6]^T$. This is a constant! The end-effector doesn't move wrt. frame {6}!



Example 4 – Step 13

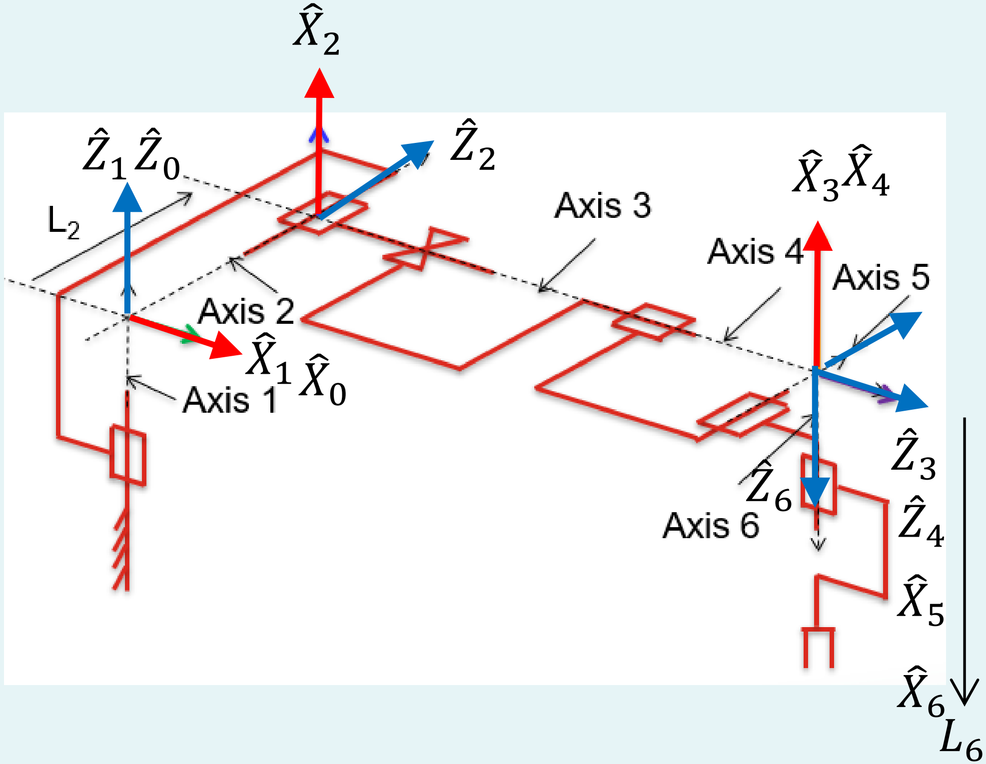
- Calculate the position and orientation of end-effector wrt. frame {0}.

$$ {}^{0}P = {}^{0}_{6}T \cdot {}^{6}P = {}^{0}_{6}T \cdot \begin{bmatrix} 0 \\ 0 \\ L_6 \\ 1 \end{bmatrix} $$

- The expression is rather large and is not typed out here.
- We can however do a sanity check by putting in numerical values into ${}^{0}_{6}T$.

Example 4 – Step 13 Continued

- In the configuration as shown:
  - $\theta_1 = 0$
  - $\theta_2 = -90$
  - $\theta_3 = 0$ (constant!)
  - $\theta_4 = 0$
  - $\theta_5 = 90$
  - $\theta_6 = 0$
- Using these values gives:
  - $$^{o}P = \begin{bmatrix} d_3 \\ L_2 \\ -L_6 \\ 1 \end{bmatrix}$$
  - Result makes sense!



Content

- Introduction to Forward Kinematics
- Denavit-Hartenberg Parameters
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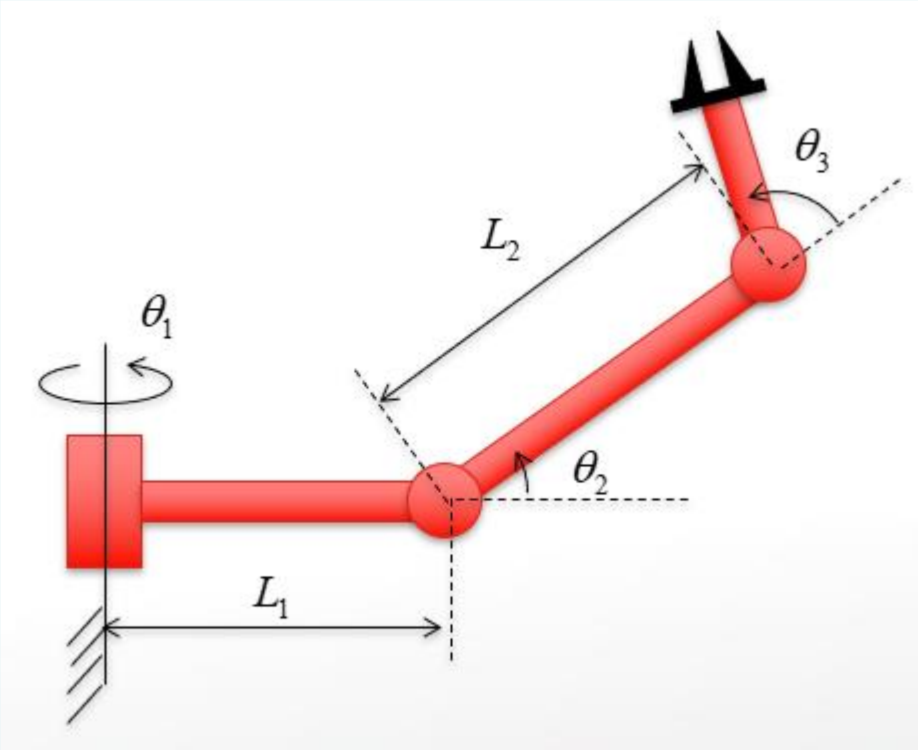
- More Examples
- Tutorial Questions

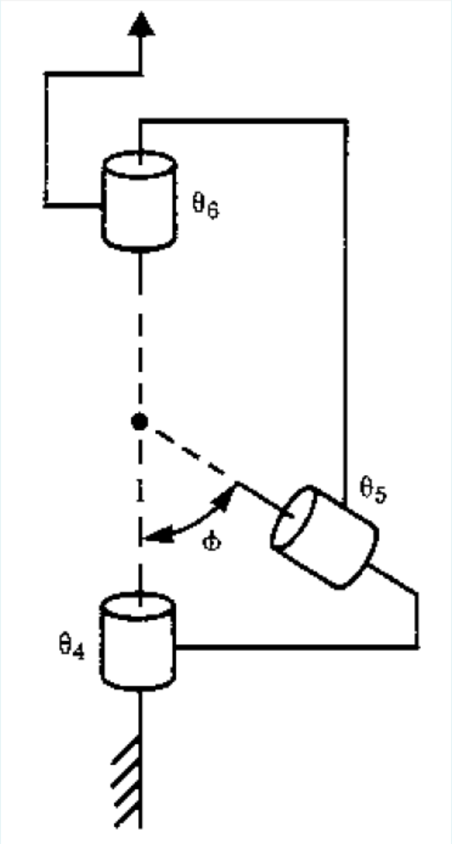
Question 1

- For the robot on the right:
  - Derive the DH parameters,
  - Sketch the frames,
  - And calculate the Transform matrix.
  - Where is the end-effector?



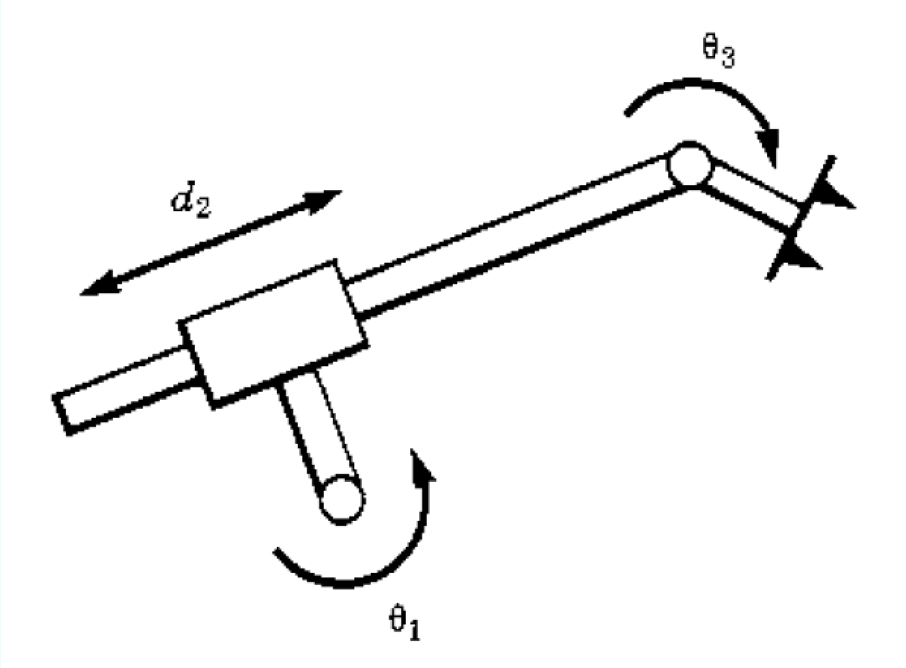
Question 2

- For the robot on the right:
  - Derive the DH parameters,
  - Sketch the frames,
  - And calculate the Transform matrix.
  - Where is the end-effector?



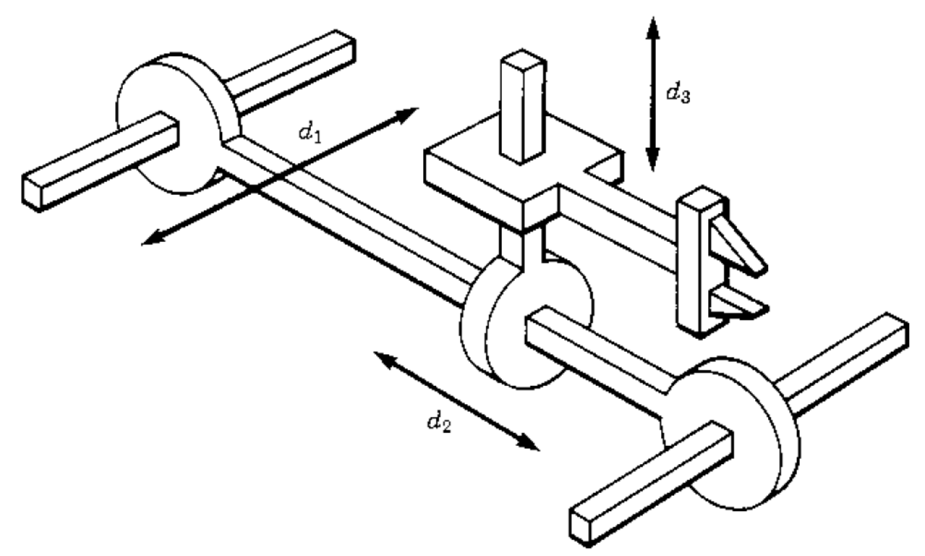
Question 3

- For the robot on the right:
  - Derive the DH parameters,
  - Sketch the frames,
  - And calculate the Transform matrix.
  - Where is the end-effector?



Question 4

- For the robot on the right:
  - Derive the DH parameters,
  - Sketch the frames,
  - And calculate the Transform matrix.
  - Where is the end-effector?



Thank you for your attention!

Any questions?